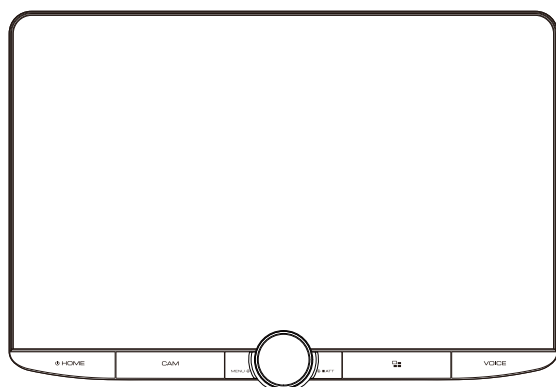


KENWOOD

SERVICE MANUAL

GPS NAVIGATION SYSTEM/MONITOR WITH RECEIVER

DMX1037S, DMX1057XR, DMX9720XDS, DMX9720XS, DNR1007XR



Made for
iPhone | iPod

Works with
Apple CarPlay

android
auto

((Siriusxm))
READY

HDMI
HIGH-DEFINITION MULTIMEDIA INTERFACE

SD
XC

HD Radio
The HD Radio receiver mode

Hi-Res
AUDIO

Hi-Res
AUDIO
WIRELESS

Hi-Res Audio logo and Hi-Res Audio
Wireless logo are used under license
from Japan Audio Society.

dab+

HD2/HD3
Bookmark
Digital Sound
Program Info
Artist Experience
Emergency Alerts

LDAC

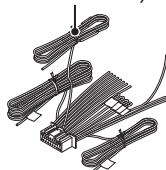
K2
TECHNOLOGY

Bluetooth

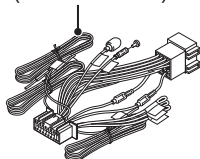
WiFi
CERTIFIED

DMX9720XDS (Europe) : E1
DMX9720XDS (Australia) : X1
DMX9720XS (Indonesia) : A1
DMX9720XS (Other Areas) : M1

DC cord
(E3A-0578-00)



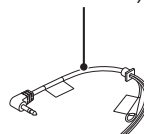
DC cord
(E3A-0579-00)



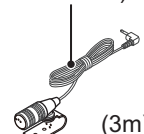
Remote ctl(IR)
(A7A-0028-00)



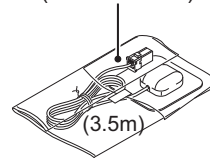
Cord with plug
(E3A-0613-00)



Microphone
(T9B-0066-00)



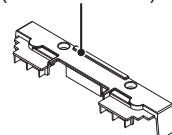
Antenna assy
(T9A-0070-00)



Cover
(F0G-0409-00) x2



Cover
(F0G-0406-00)



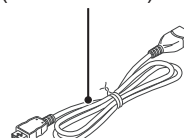
Bracket
(J2B-0592-00)



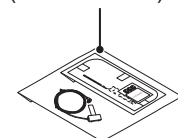
Bracket
(J2B-0596-00)



USB cord
(E3A-036x-00)



Antenna cable
(E3A-0067-00)



Special screw
(N0Z-0367-01) x6



Screw (M5x8)
(QYSSSP5008ZA) x6



Screw (M5x8)
(QYSTSP5008ZA) x6



Tap screw (M3x6)
(QYSDST3006MA) x2



LF PbF

This product uses Lead Free solder.

This product complies with the RoHS directive for the European market.

SECTION 1

PRECAUTION

1.1 Safety Precautions

- (1) This design of this product contains special hardware and many circuits and components specially for safety purposes. For continued protection, no changes should be made to the original design unless authorized in writing by the manufacturer. Replacement parts must be identical to those used in the original circuits. Services should be performed by qualified personnel only.
- (2) Alterations of the design or circuitry of the product should not be made. Any design alterations of the product should not be made. Any design alterations or additions will void the manufacturers warranty and will further relieve the manufacture of responsibility for personal injury or property damage resulting therefrom.
- (3) Many electrical and mechanical parts in the products have special safety-related characteristics. These characteristics are often not evident from visual inspection nor can the protection afforded by them necessarily be obtained by using replacement components rated for higher voltage, wattage, etc. Replacement parts which have these special safety characteristics are identified in the Parts List of Service Manual. Electrical components having such features are identified by shading on the schematics and by (Δ) on the Parts List in the Service Manual. The use of a substitute replacement which does not have the same safety characteristics as the recommended replacement parts shown in the Parts List of Service Manual may create shock, fire, or other hazards.
- (4) The leads in the products are routed and dressed with ties, clamps, tubings, barriers and the like to be separated from live parts, high temperature parts, moving parts and/or sharp edges for the prevention of electric shock and fire hazard. When service is required, the original lead routing and dress should be observed, and it should be confirmed that they have been returned to normal, after reassembling.

1.2 Warning

- (1) This equipment has been designed and manufactured to meet international safety standards.
- (2) It is the legal responsibility of the repairer to ensure that these safety standards are maintained.
- (3) Repairs must be made in accordance with the relevant safety standards.
- (4) It is essential that safety critical components are replaced by approved parts.
- (5) If mains voltage selector is provided, check setting for local voltage.

1.3 Caution

Burrs formed during molding may be left over on some parts of the chassis.

Therefore, pay attention to such burrs in the case of pre-forming repair of this system.

1.4 Critical parts for safety

In regard with component parts appearing on the silk-screen printed side (parts side) of the PWB diagrams, the parts that are printed over with black such as the resistor ($\blacksquare$), diode ($\blacksquare$) and ICP ($\bullet$) or identified by the " Δ " mark nearby are critical for safety. When replacing them, be sure to use the parts of the same type and rating as specified by the manufacturer.
(This regulation dose not Except the J and C version)

1.5 Remote control

The Lithium battery is in danger of explosion if replaced incorrectly. Replace it only with the same or equivalent type.

1.6 Preventing static electricity

Electrostatic discharge (ESD), which occurs when static electricity stored in the body, fabric, etc. is discharged, can destroy the semiconductors. Take care to prevent this when performing repairs.

1.6.1 Grounding to prevent damage by static electricity

Static electricity in the work area can destroy the optical pickup (laser diode) in devices such as laser products.

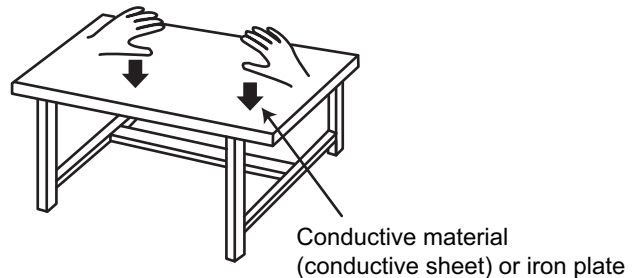
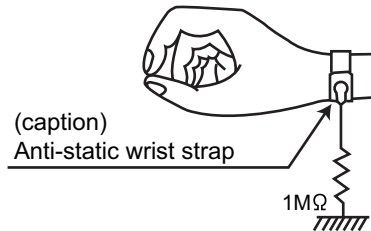
Be careful to use proper grounding in the area where repairs are being performed.

- (1) Ground the workbench

Ground the workbench by laying conductive material (such as a conductive sheet) or an iron plate over it before placing the traverse unit (optical pickup) on it.

- (2) Ground yourself

Use an anti-static wrist strap to release any static electricity built up in your body.



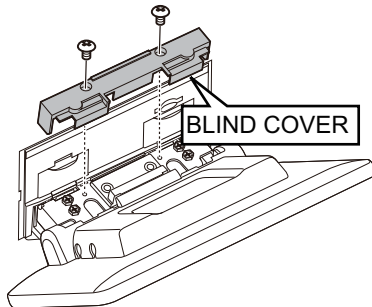
SECTION 2

SPECIFIC SERVICE INSTRUCTIONS

■BLIND COVER

NOTE:

When checking the operation or assembling, the power of the main unit will not turn on unless the accessory BLIND COVER (symbol: 362) is attached.



2.1 Precautions when replacing the service board

The following board is supplied as a set of 2 types of boards. Use them as necessary.

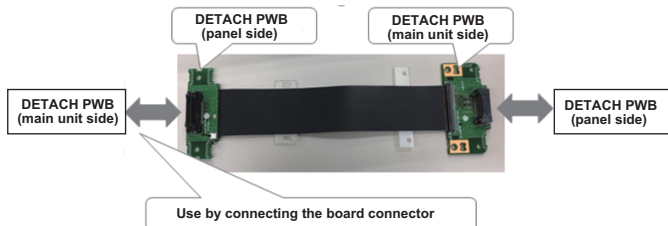
- CB-XJ8092K07N: DETACH PWB (panel side) + DETACH PWB (main unit side)

2.2 Extension cable

The extension cable is used by connecting the detached board to the panel side and the main unit side when checking the operation. Please use if necessary.

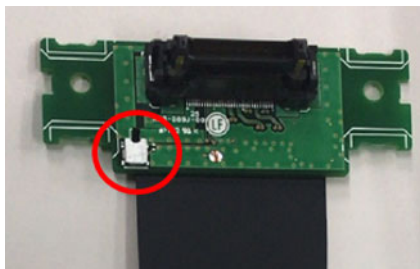
Please purchase the following 2 part numbers, assemble and use them.

- CB-XJ8092K07N: DETACH PWB (panel side) + DETACH PWB (main unit side)
- E3J-0058-00: FPC



NOTE:

- Do not use it for noise-related confirmation, as it may cause noise due to extension.
- If the mechanical switch on the DETACH PWB (PANEL SIDE) is not pressed, the power will not turn on. Therefore, check the operation with the switch pressed.



2.3 Precautions when replacing the IC

If the following IC is defective, replace the whole PWB.

Ref No.	PWB ASS'Y	Description
SN1	GYRO PWB	Replace the GYRO PWB
IC503	DIGITAL IO PWB	Replace the DIGITAL IO PWB
IC1701		
IC600	VIDEO PWB	Replace the VIDEO PWB
IC2301	AUDIO PWB	Replace the AUDIO PWB

2.4 Precautions when replacing the DIGITAL IO PWB

2.4.1 NCM MAC Address setting

When DIGITAL IO PWB is replaced, it is necessary to set the NCM MAC Address. Refer to "4.6.11 NCM MAC Address" for more information.

2.4.2 Import Gyro Adjustment Data

Import Gyro Adjustment Data if DIGITAL IO PWB or GYRO PWB was replaced. Refer to "4.6.2.1 How to import Gyro Adjustment Data" for more information.

2.4.3 Model setting

In this model, model information is stored in eMMC of SoC.

This information is not set in the service parts of the DIGITAL IO PWB.

Therefore, when replacing the DIGITAL IO PWB, it is necessary to set the model by the following procedure.

2.4.3.1 How to set the model

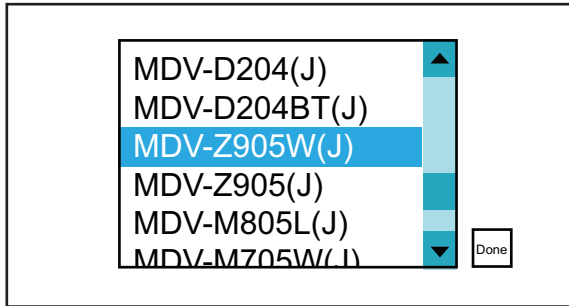
- (1) If you turn on Battery/ACC on the set with no model information set, the following screen will be displayed.



- (2) On this screen, if perform a special operation (top left of screen (1) → top right (2) → bottom left (3) → bottom right (4)), will be transition to the model selection screen.



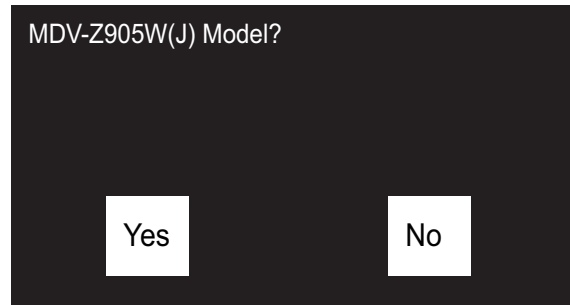
- (3) Since a list of selectable models is displayed on the model selection screen, select the model you want to set and touch "Done".



- (4) Confirm that there is no mistake in the selected model and touch "Yes".

The model is defined and the screen transition to the normal screen.

*At this time, the set does not restart. (If "No" touched, return to the model selection screen)



- (5) Since the setting values are saved in the eMMC after setting in this procedure, start up with the model you set up the next time the set is started up.

NOTE:

Depending on the combination, there is a possibility that it will not start up, so be careful to set it.

2.5 Data storage content of ICs

PWB	Ref No.	Part name	Data preservation contents (*Need to setting to after the replacement of IC)
DIGITAL IO PWB	IC1701	eMMC	User setting data Camera setting data Sound various setting data BtPhone setting data Play history data Car Play setting iPod setting data *Serial No. Audio setting data Phone book data Telephone number preset data *Production mode test result data Wallpaper customize data for DOP *File that Garmin is using (DNR model only) Pairing device information that A&W SDK is using Pairing device information managed by the BtBackEnd Log file *Garmin license file (DNR model only) Custom opening image *White balance adjustment value Image quality adjustment values Tuner Span (Area) information Tuner information *Gyro calibration data (GYRO PWB mounted model only) *USB NCM MAC address
AUDIO PWB	IC2301	MAIN uCOM	Error information DC offset information VSP (vehicle speed pulse)
VIDEO PWB	IC300	PANEL uCOM	*VCOM data *Key illumination white color adjustment value

2.6 How to repair a fuse pattern

2.6.1 Purpose of fuse pattern

In order to prevent serious damage on the circuit, fuse pattern is prepared on the GND line of RCA Terminal. This damage may take due to improper part replacement with a external equipment via RCA line.

2.6.2 Repair Procedure

- (1) Check the shorted circuit at the meltdown point.
Need to clean up if the shorted circuit or carbonization happen at the fuse pattern.
- (2) Add following part on the fuse pattern.
- (3) Check output level.

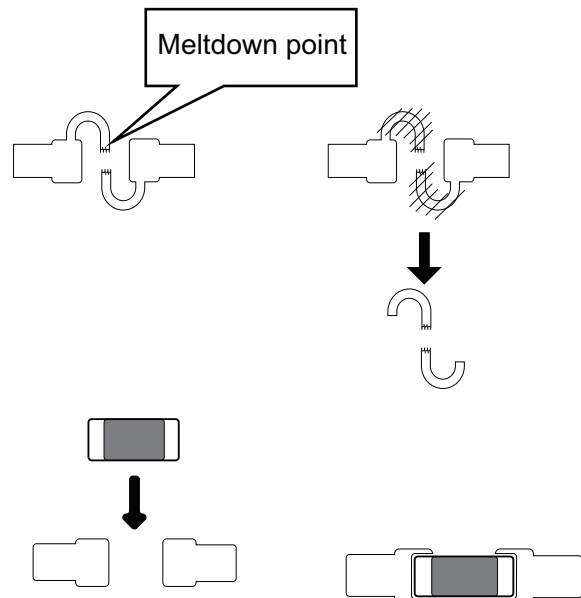
2.6.3 After finished repair

Due to improper part replacement, this meltdown occurs.

Thus please notice following information when the unit is returned to your customer.

Things to be checked before installing the unit.

- (1) Check the GND line of external amplifier or other equipment which must connect properly.
- (2) Check whether the GND line is not short-circuited with the battery terminal. (do not short-circuit these lines)



Part Number	Part Name	SPEC
F53-0513-08	PATTERN FUSE	4A

2.7 Components description

2.7.1 GYRO PWB (XJ5-009)

Ref. No.	Application / Function	Operation / Condition / Compatibility
SN1	Inertial Sensor	6-axis sensor

2.7.2 AUDIO PWB (XJ6-153)

Ref. No.	Application / Function	Operation / Condition / Compatibility
IC1	Multi channel Voltage Regulator	ILM8V,CD8V, A8.5V, SW5V, BU5V, P-CON, P-ANT Power Supply
IC31	Comparator IC	ACC Detect
IC41	Voltage Regulator	BU3.3V
IC101	Power IC	Power AMP
IC115	OR Logic IC	PWIC_BEEP_Level_shift
IC201	Electronic Vol IC	Audio Vol Control, Audio Signal Select
IC202	operational amplifier	Preout mix for noise cancelling in Audio DSP
IC203	operational amplifier	Preout mix for noise cancelling in Audio DSP
IC251	operational amplifier	-1.6dB attenuate audio signal for SXM/ADS
IC252	operational amplifier	-1.6dB attenuate audio signal for SXM/ADS
IC261	Isolation amplifier	ISO Amp for SXM/ADS
IC290	Analog Multiplexer/ Demultiplexer	Audio Selector for SXM/ADS
IC302	AM/FM Radio receiver, HD Radio tuner	AM/FM Radio receiver, HD Radio tuner
IC303	Serial flash memory	Software for IC302
IC304	Buffer IC	Buffer for I2S of AM/FM Radio
IC305	Buffer IC	Buffer for ZIF of AM/FM Radio
IC306	I2C Bus Repeater	I2C Bus Repeater for AM/FM Radio
IC307	Buffer IC	Buffer for ZIF of AM/FM Radio
IC351	Voltage Regulator	Tuner1.8V
IC352	Voltage Regulator	Tuner3.3V
IC511	OR Logic IC	SXM_TX Level shift
IC512	OR Logic IC	SXM_RX Level shift
IC590	Bus Buffer IC	Buffer for TV_Remote_Out
IC601	Line amp IC	5V Preout
IC851	Comparator IC	Vehicle speed pulse Detect
IC852	OR Logic IC	Selector for Vehicle speed pulse
IC1101	operational amplifier	audio buffer to PWIC
IC1102	operational amplifier	audio buffer to PWIC
IC2001	Comparator	BU voltage detect
IC2301	Micro Computer	Slave CPU of Telechips, Control of Power, XJ6, etc.
IC2302	Voltage Detect IC	RESET control

Ref. No.	Application / Function	Operation / Condition / Compatibility
IC2303	Comparator	RESET control
IC2304	AND GATE IC	Remote control input
IC2501	DSP	AUDIO DSP, DAC
IC2502	Audio DAC	Subwoofer out
IC2503	Audio DAC	Alter / NAVI out
IC2507	Regulator	Regulator IC for A5V
IC2508	Regulator	Regulator IC for A3.3V
IC2601	OP Amp	MIC Amp
IC2602	Regulator	Regulator IC for MIC5V
Q11	Transistor	Switch for SW5V
Q12	Digi Transistor	SW5v control
Q51	Transistor	Illumi Detect
Q60	Transistor	Camera control
Q61	Transistor	Camera control
Q62	Transistor	Camera control
Q63	Transistor	Camera control
Q80	Transistor	Parking Detect
Q81	Transistor	Parking Detect
Q82	Transistor	Parking Detect
Q90	Digi Transistor	Reverse Detect
Q151	Digi Transistor	Preout mute drive
Q152	Transistor	Preout mute drive
Q155	Digi Transistor	Preout mute drive
Q156	Transistor	Preout mute drive
Q161	Digi Transistor	Preout FL mute
Q162	Digi Transistor	Preout FR mute
Q163	Digi Transistor	Preout RL mute
Q164	Digi Transistor	Preout RR mute
Q165	Digi Transistor	Preout SWL mute
Q166	Digi Transistor	Preout SWR mute
Q191	Transistor	AVOUT mute drive
Q193	Digi Transistor	AVOUT AL mute
Q194	Digi Transistor	AVOUT AR mute
Q2004	Transistor	BU voltage detect (Hi voltage)
Q2301	Transistor	SW3.3V control
Q2302	Transistor	Q301 control
Q2303	Transistor	u-COM RESET control
Q2304	Transistor	u-COM RESET control
Q2305	Transistor	u-COM RESET control
Q2306	Transistor	DET_MUTE control
Q2307	Transistor	u-COM RESET control
Q2308	Transistor	u-COM RESET control
Q2309	Transistor	u-COM RESET control
Q2310	Transistor	C325-C330 charge
Q2311	Transistor	Q310 control
Q2312	Transistor	PWIC MUTE control

Ref. No.	Application / Function	Operation / Condition / Compatibility
Q2313	Transistor	PWIC MUTE control
Q2314	Transistor	DET_MUTE control
Q2401	Transistor	DET_MUTE control
Q2402	Transistor	Panel_Det_control
Q2501	Transistor	DSP power down control
Q2502	Transistor	DAC RESET control

2.7.3 VIDEO PWB (XJ8-092)

Ref. No.	Application / Function	Operation / Condition / Compatibility
IC1	Remote control receiver	Remote control receiversignal output
IC100	Voltage Regulator	Panel 3.3V Regulator
IC300	Panel u-com	Processor
IC301	RESET IC	RESET Voltage detect 2.8V
IC302	OR LOGIC IC	Detection of KEY1/KEY2 and HOME
IC303	Schmitt Buffer IC	Separation from main unit side circuit
IC500	DCDC Control IC	LED POWER 21.6V DCDC converter
IC600	Touch Panel Controller IC	Touch Panel Controller
IC601	Voltage Regulator	Touch Panel 3.3V Regulator
IC700	LCD Back Light Controller IC	LCD Back Light Control
Q30	Transistor	Key Illumi LED Control
Q31	Transistor	Key Illumi LED Control
Q32	Transistor	Key Illumi LED Control
Q50	Transistor	Key Illumi LED Control
Q51	Transistor	Key Illumi LED Control
Q52	Transistor	Key Illumi LED Control
Q300	Digi Transistor	Reset Voltage Out
Q301	Digi Transistor	Reset Voltage Out

2.7.4 DIGITAL IO PWB (XJA-080)

Ref. No.	Application / Function	Operation / Condition / Compatibility
A251	BT+WLAN combo module	Bluetooth + W-LAN combo module
IC21	LCD Controller	Video scaler and timing controller
IC22	AND Logiv IC	Panel reset control
IC81	comparator IC	Panel detachment detection
IC111	Regulator	VIDEO selector IC V5V Regulator IC
IC112	Video SW IC	Composite signal selector
IC151	Vedeco encoder IC	BT.656 to CVBS OUT MACROVISION
IC231	USB Charging Port Controller	USB charging port control
IC251	Regulator	BT+1.8V Regulator IC

Ref. No.	Application / Function	Operation / Condition / Compatibility
IC252	Regulator	BT+3.6V Regulator IC
IC301	RF IC	DAB TUNER RF IC
IC302	Regulator	DAB RF IC DAB1.2V Regulator IC
IC303	ANT detection IC	DAB ANT short detection
IC402	Regulator	HDMI1.2V Regulator
IC451	Bidirectional multiplexer IC	HDMI & HDTV I selector IC
IC501	Regulator	HDTV I1.2V Regulator IC
IC502	HDTV I IC	HDTV I input to BT.656 Output
IC701	Inverter IC	Transform 3.3V→5V for FAN PWM
IC721	AND GATE IC	Transform 3.3V→5V for DASH CAM I/F
IC722	3state Buffer IC	Buffer for DASH CAM I/F
IC723	AND GATE IC	Transform 5V→3.3V for DASH CAM I/F
IC781	3state Buffer IC	Buffer for iDATA LINK I/F
IC782	3state Buffer IC	Transform 5V→1.8V for iDATA LINK I/F
IC831	Audio DAC	2 zone out
IC853	Low Noise Amp	GNSS Low Noise Amp GaAs MMI IC
IC854	GNSS/GPS Receiver IC	GNSS/GPS Receiver
IC855	Comparator	GNSS/GPS Antenna detect
IC856	Comparator	GNSS/GPS Antenna detect
IC857	Regulator	Regulator IC for GANT3.3V
IC858	Regulator	Regulator IC for G3.3V
IC901	OP Amp	Iso. Amp for AVIN audio
IC902	OP Amp	Iso. Amp reference voltage output
IC1001	SoC	Processor for Digital Media
IC1060	Triple Non-Inverter	Overvoltage protection IC
IC1151	AND Logiv IC	TCON/PANEL PCB/HDTV I/HDMI RESET
IC1152	RTC	Real time clock module
IC1153	Triple Non-Inverter	Overvoltage protection IC
IC1154	Triple Non-Inverter	Overvoltage protection IC
IC1601	16Gb x32bit LPDDR4 memory	For SoC memory
IC1701	16GB or 8GB eMMC memory	For SoC memory
IC1751	256Mb Serial nor Flash memory	For SoC memory
IC1801	iPod Authentication Coprocessor	iPod Authentication
IC2001	DCDC controller IC	SOC4.3V DCDC converter
IC2101	Power Management IC	SOC Power Management IC

Ref. No.	Application / Function	Operation / Condition / Compatibility
IC2151	Regulator IC	eMMC1.8V Regulator IC
IC2301	DCDC controller IC	D3.3V DCDC converter
IC2401	DCDC controller IC	D5.0V DCDC converter
IC2501	DCDC controller IC	USB5.0V DCDC converter
IC2502	Current-Shunt Momitor	USB VBUS current detect
IC2701	Regulator IC	SD3.3V Regulator IC
IC2702	Regulator IC	TCON1.5V Regulator IC
IC2703	High-Side Power Switc	PANEL14V High-Side Power Switc
Q1	Transistor	LCD STANDBY Control for HD LCD
Q21	Transistor	Emitter follower circuit of VIDEO line
Q81	Transistor	Panel detachment detection
Q111	FET	Video selector I2C SCL Lvel shift (3.3V→5V)
Q112	FET	Video selector I2C SDA Lvel shift (3.3V→5V)
Q113	Transistor	V5V control for IC111
Q114	Transistor	V5V control for IC111
Q401	FET	HDMI DDC_SCL control
Q741	Transistor	STE REMO Pull up control
Q742	FET	STE REMO Pull up SW
Q743	FET	STE REMO Pull up SW
Q801	FET	GNSS/GPS Antenna power supply
Q1151	FET	ADS_TX Lvel shift (1.8V→3.3V)
Q1153	Transistor	SOC RESET
Q1202	Transistor	WiFi RESET Level shift (1.8V→3.3V)
Q1203	Transistor	WiFi RESET Level shift (1.8V→3.3V)
Q1801	Transistor	ipod CP I2C SDA control
Q1802	Transistor	ipod CP I2C SCL control
Q1803	Transistor	ipod CP I2C control
Q1804	Transistor	ipod CP Power control
Q1805	Transistor	ipod CP Power control
Q2101	Transistor	Power Management IC discharge control
Q2102	FET	Power Management IC discharge control (IO1.8V)
Q2103	FET	Power Management IC discharge control (IO3.3V/eMMC3.3VV)
Q2501	Transistor	VBUS2 over current detect
Q2502	Transistor	VBUS2 over current detect
Q2503	Transistor	VBUS2 over current detect out
Q2504	Transistor	VBUS2 over current detect out

2.8 Microcomputer's terminal description

2.8.1 Main micro computer (AUDIO PWB: IC2301)

Pin No.	Pin Name	I/O	Application	Processing/Operation/Description
1	VP_TO_M_DAT	I	Sensor data input	
2	M_TO_VP_CLK	O	Sensor clock output	
3	POW_VF_CTL0	O	DC-DC_IC frequency control	
4	POW_VF_CTL1	O	DC-DC_IC frequency control	
5	EXTAD_CS0	O	Sensor (gyro) CS0	
6	MUTE_AMP	O	Power IC voice mute control	H: Unmute, L: Mute
7	POW_AMP (PWIC_STBY)	O	Power IC standby control	H: Normal, L: Standby
8	PWIC_CDET	I	PWIC earth fault detection	H: Normal, L: Earth fault detection
9	STEREMO_SW	O	Steering remote control pull-up constant switching output	H: Pull up 460 Ω , L: Pull up 4.7k Ω
10	VDD	-	VDD	
11	VDDA	-	VDDA	
12	VREFH	-	VREFH	
13	VREFL	-	VREFL	
14	VSS	-	VSS	
15	EXTAL	-	Oscillator connection for main system clock (16MHz)	
16	XTAL	-	Oscillator connection for main system clock (16MHz)	
17	AUDIO_SEL2	O	Audio selector CH switching (example: ADS/SXM switching)	L: External input device 3, H: External input device 4
18	DSP_CS	O	DSP Chip Select (MAIN to DSP)	
19	DAC_CS	O	DAC chip select (MAIN to DAC)	
20	DSP_DAC_BEEP_PDN	O	DSP power down/front DAC power down/BEEP_IC power down	H: Power down, L: Normal
21	D_TO_M_DAT	I	DSP data communication (DSP to MAIN) / DAC data communication (DAC to MAIN)	
22	M_TO_D_CLK	O	DSP clock communication (MAIN to DSP)/DAC clock communication (MAIN to DAC)	
23	SW14_PGOOD	I	SW14V abnormality monitoring	H: SW14V normal, L: SW14V abnormal
24	FAN_PWM	O	FAN rotation speed control	PWM output 32k Duty 50%
25	AUDIO_SEL1	O	Audio selector CH switching (example: AVIN/AUX (F-AUX) switching)	L: External input device 1, H: External input device 2
26	PMIC_EN	O	PMIC power control	H: ON sequence start, L: OFF sequence start
27	SW_ILLUMI_NAVI	O	Illumination detection output at startup	L: Illumination ON, H: Illumination OFF
28	M_TO_D_DAT	O	DSP data communication (MAIN to DSP)/DAC data communication (MAIN to DAC)	
29	CAN_RESET	O	CAN uCom RESET	H: Reset, L: Reset release
30	M_TO_C_STBY_REQ	O	CAN microcomputer STANDBY request	H: Normal, L: Standby request
31	M_TO_C_DAT	O	CAN UART for communication between micro-computers	
32	C_TO_M_DAT	I	CAN UART for communication between micro-computers	
33	EVOL_MUTE_S	O	E-VOL SUB mute	H: Unmute, L: Mute

Pin No.	Pin Name	I/O	Application	Processing/Operation/Description
34	DAC_PDN	O	DAC Power Down (Common for Mix path/For SubWoofer/2Zone)	H: Power down, L: Normal
35	CAN_WDT_EN	O	CAN WDT Enable	H: WDT disabled, L: WDT enabled
36	A_POW_PWM	O	5VPREOUT_IC frequency control	
37	VSS	-	VSS	
38	VDD	-	VDD	
39	SW_ILLMI	I	Illumination SW	H: ILLMI OFF, L: ILLMI ON
40	SW_PARKING	I	Parking SW	H: PARKING ON, L: PARKING OFF
41	SW_REVERSE	I	Reverse gear SW	H: REVERSE OFF, L: REVERSE ON
42	PS2_BAT	I	Back-up power supply detection	H: Normal, L: Instantaneous drop detection
43	AD_HWCHK	I	Hardware check mode input	H: Normal, L: Transition to test mode
44	AD_KEY1	I	Steering remote control key input 1	
45	AD_KEY2	I	Steering remote control key input 2	
46	AD_TILTDET	I	Tilt mechanism (voltage type) position detection input	
47	AD_DCERR	I	Power amplifier output DC potential difference detection	
48	LINE_MUTE	I	LINE-MUTE input detection	
49	C_TO_M_WAKEUP	I	For returning to low power consumption from a CAN microcomputer	H: WAKEUP, L: Normal
50	KEY_INPUT	I	HOME key	H: Normal, L: Press HOME key
51	PS1_ACC	I	ACC detection	H: ACC ON, L: ACC OFF
52	IGN_DET	I	IGN detection	
53	ETC_TXD	O	ETC/DashCam communication production (Auto-line/CPRM writing) communication (UART data transmission line)	
54	ETC_RXD	I	ETC/DashCam communication production (Auto-line/CPRM writing) communication (UART data reception line)	
55	CAMERA_TXD	O	CMOS camera communication (UART TX)	
56	CAMERA_RXD	I	CMOS camera communication (UART RX)	
57	AD_KEY3	I	Steering/remote control key input 3	
58	SW_REVERSE_NAVI	O	Reverse detection output at startup	L: Reverse ON, H: Reverse OFF
59	NAVI_NMIPDN	O	Navi power down	H: Normal, L: Power down
60	VSS	-	VSS	
61	VDD	-	VDD	
62	N_TO_M_RDY	I	NAVI data transmission/reception permission signal (NAVI to MAIN)	
63	M_TO_N_REQ	O	NAVI data transmission request (MAIN to NAVI)	
64	M_TO_N_DAT	O	NAVI data communication (MAIN to NAVI)	
65	N_TO_M_DAT	I	NAVI data communication (NAVI to MAIN)	
66	M_TO_N_CLK	O	NAVI clock communication (MAIN to NAVI)	
67	N_TO_M_REQ	I	NAVI data reception request (NAVI to MAIN)	
68	NAVI_RST	O	NAVI reset (MAIN to NAVI)	H: Reset, L: Reset release
69	PRE_MUTE_SW	O	Pre-out mute (for subwoofer)	H: Mute, L: Unmute
70	PRE_MUTE_FR	O	Pre-Muted (For Frontier)	H: Mute, L: Unmute
71	WREMO_J	I	Steering remote control input	

Pin No.	Pin Name	I/O	Application	Processing/Operation/Description
72	I2C_SCL	O	EVOL communication (ADDR: 0x88) BEEP_IC communication (ADDR: 0x0A)	
73	I2C_SDA	I/O	EVOL communication (ADDR: 0x88) BEEP_IC communication (ADDR: 0x0A)	
74	EVOL_MUTE2	O	E-VOL SW Mute	H: Unmute, L: Mute
75	EVOL_MUTE1	O	E-VOL rear mute	H: Unmute, L: Mute
76	EVOL_MUTE0	O	E-VOL Front Mute	H: Unmute, L: Mute
77	VSP_IN	I	Speed pulse input	
78	VSP_DA	O	Speed pulse input switching (digital/analog)	H: Digital, L: Analog
79	POW_REG2	O	For conventional power supply control without PMIC	H: POWER ON, L: POWER OFF
80	SI_CTL	O	Security indicator LED	
81	REMO_IN	I	REMO/KW: Wired REMO input	
82	PM_MINUS	O	Tilt mechanism (voltage type) PM-	DC control, PWM reserved
83	PM_PLUS	O	Tilt mechanism (voltage type) PM+	DC control, PWM reserved
84	PM_SPEED	O	Tilt mechanism (voltage type) speed setting	Control LDO output for tilt, H: Slow (VM=4V), L: Fast (VM=7V)
85	TOUCH_TV_REMO	O	REMO OUT at General-purpose TV CTRL touch panel operation (waveform generation in timer)	
86	VSS	-	VSS	
87	VDD	-	VDD	
88	POW_ON	O	System regulator control (P-ON)	H: POWER ON, L: POWER OFF
89	POW_REG (SW14V_PON)	O	SW14V ON/OFF	H: SW14V ON, L: SW14V OFF
90	PMIC_PGOOD	I	PMIC power abnormality monitor input	(Reserved), H: When PMIC is normal, L: When PMIC is abnormal
91	BEEP	O	BEEP PWM output	
92	JTAG_TDO/ noetm_TRACE_SWO	O	JTAG: Test Data Output SWD: trace data output (SWO)	
93	POW_CON	O	System regulator control (P_CON)	H: P_CON ON, L: P_CON OFF
94	POW_ANT	O	System regulator control (P-ANT)	H: P_ANT ON, L: P_ANT OFF
95	JTAG_TDI	I	JTAG: Test Data Input	
96	JTAG_TCLK/ SWD_CLK	I	Clock into the core	
97	RESET_b	I	Reset	
98	JTAG_TMS/SWD_DIO	O	JTAG: Test Mode Select SWD: data in/out	
99	EXTAD_CS1	O	Sensor (G sensor) CS1	
100	M_TO_VP_DAT	O	Sensor data output	

2.8.2 Panel micro computer (VIDEO PWB: IC300)

Pin No.	Pin Name	I/O	Application	Processing/Operation/Description
1	NC	O	Not used (L fixed)	
2	NC	O	Not used (L fixed)	
3	NC	O	Not used (L fixed)	
4	LCD_18V	O	LCD power control 2 (LCD_ENA_18 V)	Except for 6.8inch XGA model
4	NC	O	Not used (L fixed)	6.8inch XGA model
5	LCD_-6V	O	LCD power control 1 (LCD_ENA_ - 6 V)	Except for 6.8inch XGA model
5	NC	O	Not used (L fixed)	6.8inch XGA model
6	/RESET	I	System reset input	
7	VOFF	O	VOFF	
8	FLMD0_CTRL	O	FLMD0 pin control (for self programming)	
9	FLMD0	I	Flash programming mode pull-in	
10	EX_X2	O	For flash writer	
11	NC	O	Not used (L fixed)	
12	REGC	-	Regulator output stable capacitance connection for internal operation	
13	VSS	-	GND for other than the P121 to P124 and port section	
14	EVSS	-	GND of ports section other than P20 to P27 and P121 to P124	
15	VDD	-	Positive power supply for other than the P121 to P124 and port section	
16	EVDD	-	Positive power supply of ports section other than P20 to P27 and P121 to P124	
17	VF_SW2	I	BOOST SYNC Frequency switching	For rear camera high-speed starting
18	VF_SW	I	BOOST SYNC Frequency switching	For rear camera high-speed starting
19	/P_TO_N_REQ	O	Transmission request to NAVI of communication between microcomputers	
20	/N_TO_P_REQ	I	Clock output request from NAVI of communication between microcomputers	
21	BL_PWM	O	PWM output for brightness change of LCD backlight (BL_PWM)	
22	NC	O	Not used (L fixed)	
23	NC	O	Not used (L fixed)	
24	LCD_9.2V	O	LCD power control 3 (LCD_ENA_9.2 V)	Except for 6.8inch XGA model
24	NC	O	Not used (L fixed)	6.8inch XGA model
25	V-COM1	O	V-COM voltage adjustment 1	Except for 6.8inch XGA model
25	NC	O	Not used (L fixed)	6.8inch XGA model
26	V-COM2	O	V-COM voltage adjustment 2	Except for 6.8inch XGA model
26	NC	O	Not used (L fixed)	6.8inch XGA model
27	V-COM3	O	V-COM voltage adjustment 3	Except for 6.8inch XGA model
27	NC	O	Not used (L fixed)	6.8inch XGA model
28	V-COM4	O	V - COM voltage adjustment 4	Except for 6.8inch XGA model
28	NC	O	Not used (L fixed)	6.8inch XGA model
29	BOOST_H	O	LCD backlight (BOOST_ENA)	
30	NC	O	Not used (L fixed)	
31	NC	O	Not used (L fixed)	
32	NC	O	Not used (L fixed)	

Pin No.	Pin Name	I/O	Application	Processing/Operation/Description
33	NC	O	Not used (L fixed)	
34	R_VOL+	I	Rotary VOL	DNX6.2 inch model, VW model
34	NC	O	Not used (L fixed)	Except for DNX6.2 inch model, VW model
35	R_VOL-	I	Rotary VOL	DNX6.2 inch model, VW model
35	NC	O	Not used (L fixed)	Except for DNX6.2 inch model, VW model
36	MENU	I	MENU KEY	
37	KEY_ILL	O	Red illumi dimming for eject	VW model
37	NC	O	Not used (L fixed)	Except for VW model
38	/N_TO_P_RDY	I	State notification from NAVI of communication between microcomputers	
39	KEY_PWM_R	O	Key illumi PWM [RED]	Except for KW-V 6.8 inch model
39	NC	O	Not used (L fixed)	KW-V 6.8 inch model
40	KEY_PWM_G	O	Key illumi PWM [GREEN]	Except for KW-V 6.8 inch model
40	/DIMMER_ON	O	Key illumi PWM Dimmer	L: Dimmer_ON, H: Dimmer_OFF KW-V 6.8 inch model
41	KEY_PWM_B	O	Key illumi PWM [BLUE]	Except for KW-V 6.8 inch model
41	KEY_ILL_ON	O	Key illumi ON	H: Key illumi_ON, L: Key illumi_OFF KW-V 6.8 inch model
42	RXD	O	For communication with flash writer	
43	TXD	O	For communication with flash writer	
44	P_TO_N_DAT	O	Data transmission to microcontroller communication to NAVI	
45	N_TO_P_DAT	I	Receive data from NAVI of communication between microcomputers	
46	P_TO_N_CLK	O	Clock signal to NAVI of communication between microcomputers	
47	AVREF	-	Analog reference voltage input	
48	AVSS	-	Analog GND	
49	NC	O	Not used (L fixed)	
50	KEY2	I	Tact switch voltage input 2	
51	KEY1	I	Tact switch voltage input 1	
52	NC	O	Not used (L fixed)	
52	KEY3	I	Tact switch voltage input 3	
53	XDATA_U	I	Touch panel X axis voltage input (VDD side)	1 point touch resistance film model
53	XDATA_U	I	Touch panel identification	Except for 1 point touch resistance film model
54	XDATA_D	I	Touch panel X axis voltage input (GND side)	1 point touch resistance film model
54	XDATA_D	I	Touch panel identification	Except for 1 point touch resistance film model
55	YDATA_U	I	Touch panel Y axis voltage input (VDD side)	1 point touch resistance film model
55	YDATA_U	I	Touch panel identification	Except for 1 point touch resistance film model
56	YDATA_D	I	Touch panel Y axis voltage input (GND side)	1 point touch resistance film model
56	YDATA_D	I	Touch panel identification	Except for 1 point touch resistance film model
57	X0	O	Touch panel X axis control	1 point touch resistance film model
57	NC	O	Not used (L fixed)	Except for 1 point touch resistance film model
58	X1	O	Touch panel X axis control	1 point touch resistance film model
58	NC	O	Not used (L fixed)	Except for 1 point touch resistance film model
59	Y0	O	Touch panel Y axis control	1 point touch resistance film model
59	NC	O	Not used (L fixed)	Except for 1 point touch resistance film model

Pin No.	Pin Name	I/O	Application	Processing/Operation/Description
60	Y1	O	Touch panel Y axis control	1 point touch resistance film model
60	NC	O	Not used (L fixed)	Except for 1 point touch resistance film model
61	BOOST_SYNC	O	BOOST_SYNC	
62	TP_PRESS	O	Touch panel control	1 point touch resistance film model
62	NC	O	Not used (L fixed)	Except for 1 point touch resistance film model
63	PULL_UP_ON	O	Touch panel control	1 point touch resistance film model
63	NC	O	Not used (L fixed)	Except for 1 point touch resistance film model
64	NC	O	Not used (L fixed)	

SECTION 3 DISASSEMBLY

3.1 Main body

3.1.1 Removing the HINGE ASS'Y (See Fig.1 to 5)

- (1) Remove the 4 screws.

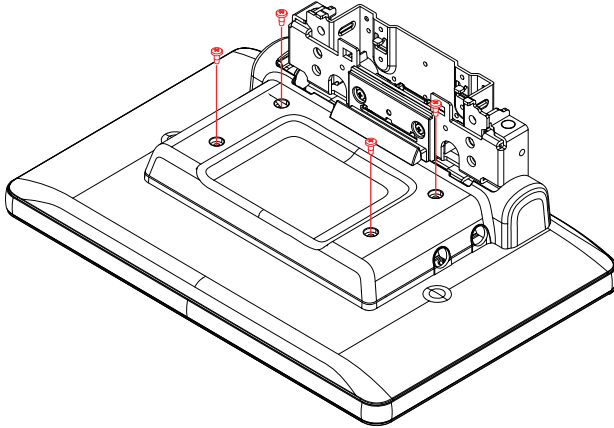


Fig.1

- (2) Lift the bottom of the hinge cover in the direction of the arrow to remove it. (See Fig.2)

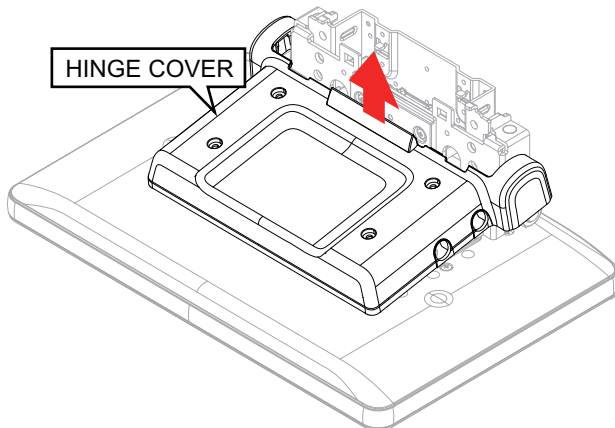


Fig.2

- (3) Remove the 4 screws. (See Fig.3)

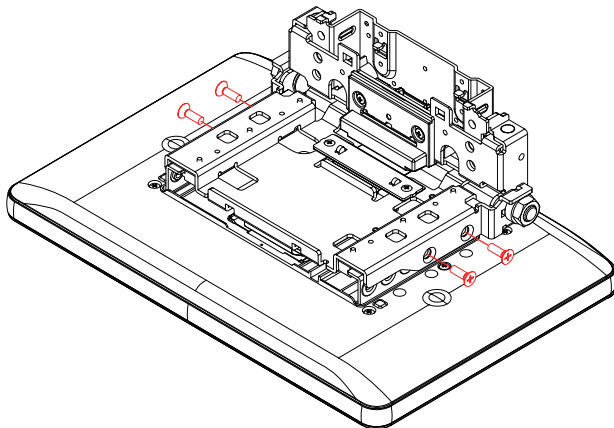


Fig.3

- (4) Slide the HINGE ASS'Y halfway in the direction of the arrow. (See Fig.4)

- (5) Remove the 2 screws, and remove the FPC from the connectors. (See Fig.4)

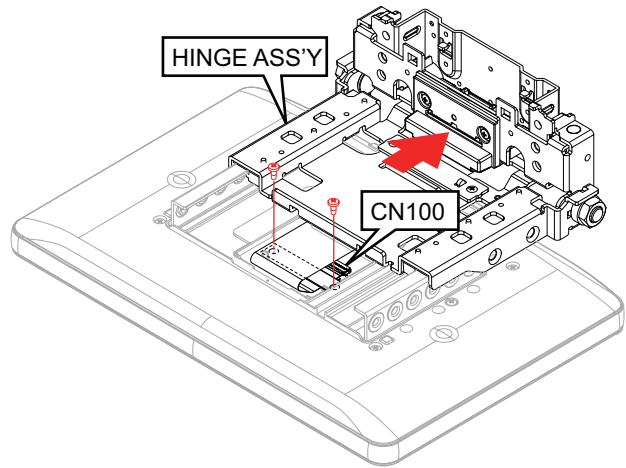


Fig.4

- (6) Slide the HINGE ASS'Y in the direction of the arrow to remove it. (See Fig.5)

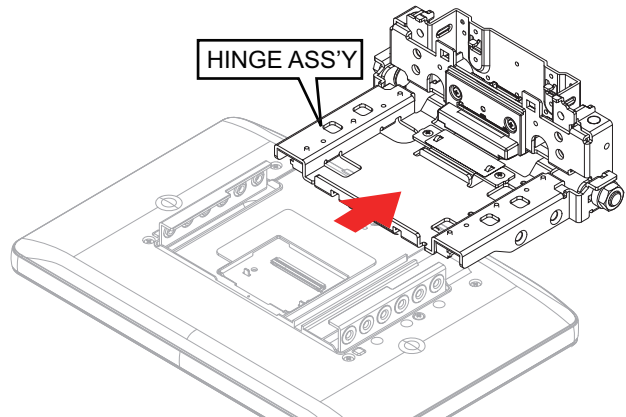


Fig.5

3.1.2 Removing the DETACH PWB (See Fig.6 and 7)

- (1) Remove the 2 screws, and remove the CONNECTOR COVER. (See Fig.6)

- (2) Disconnect the FPC from the connector. (See Fig.6)

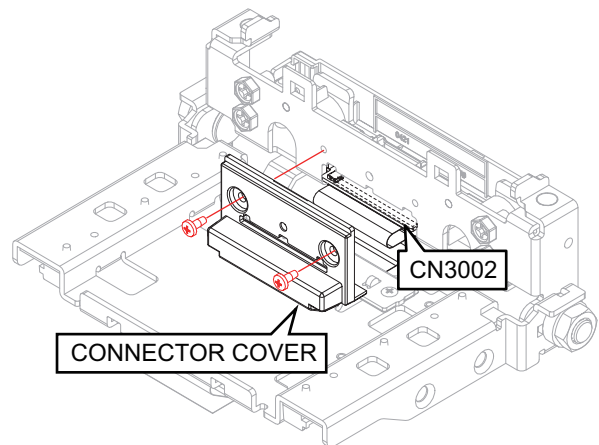


Fig.6

- (3) Remove the 2 screws, and remove the CONNECTOR COVER and DETACH PWB. (See Fig.6)

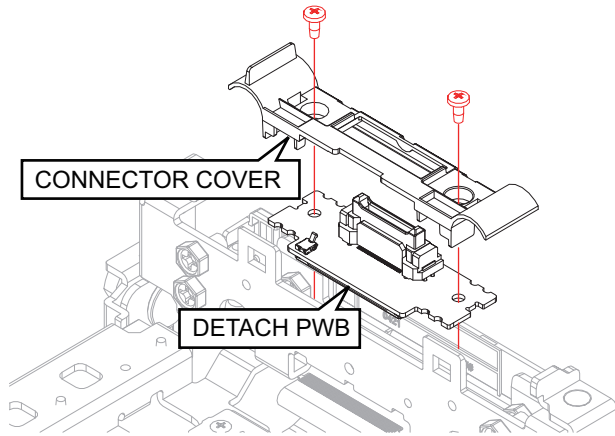


Fig.7

3.1.3 Removing the PANEL PWB (See Fig.8)

- (1) Remove the 7 screws (a), and remove the REAR COVER. (See Fig.6)
- (2) Disconnect the FPC and FFC from the connectors. (See Fig.6)
- (3) Remove the 4 screw (b), and remove the PANEL PWB. (See Fig.6)

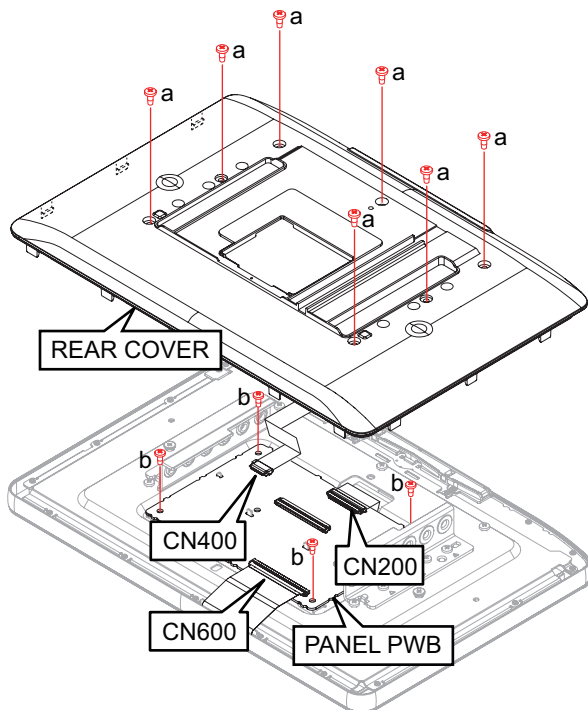


Fig.8

3.1.4 Removing the SWITCH PWB (See Fig.9)

- (1) Remove the 17 screws (a,b), and remove the BRACKET.
- (2) Remove the VOLUME KNOB.
- (3) Remove the 2 screws (c), and remove the LIGHTING BOARD.
- (4) Remove the 7 screws (d), and remove the SWITCH PWB.

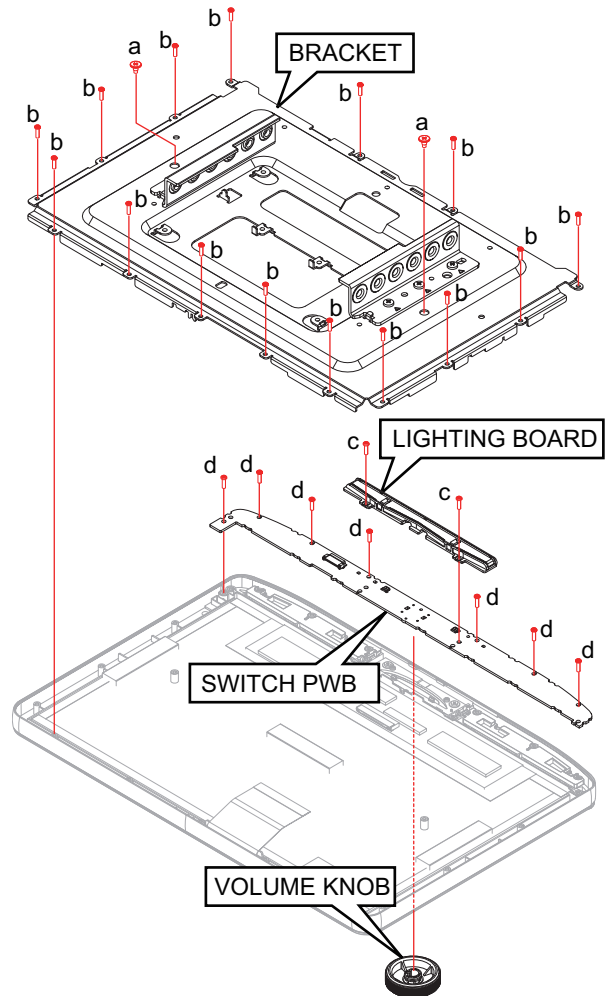


Fig.9

3.1.5 Removing the SD PWB (See Fig.10 to 14)

- (1) Remove the 6 screws, and remove the TOP PLATE. (See Fig.10)

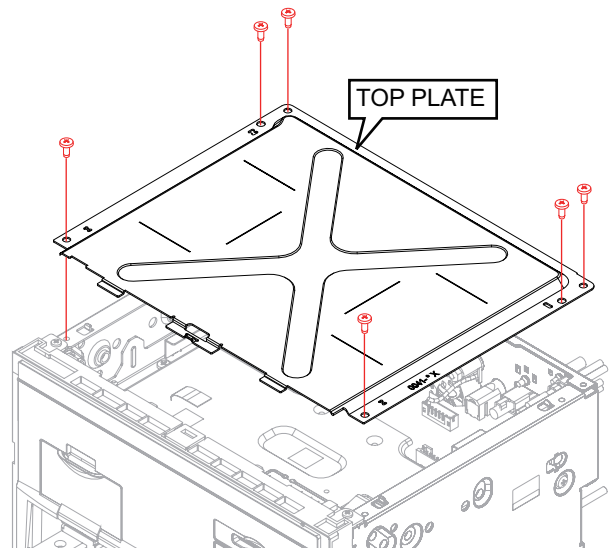


Fig.10

- (2) Remove the COVER. (See Fig.11)
- (3) Remove the 6 screws (a,b). (See Fig.11)

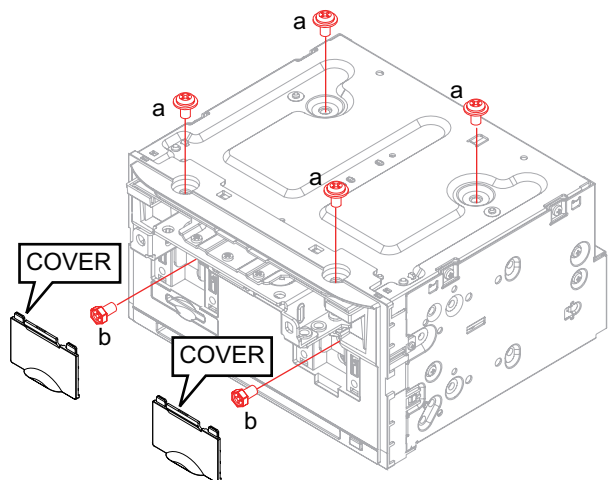


Fig.11

- (4) Pull the panel arm toward you, remove the 2 screws (a), and remove the BRACKET. (See Fig.12)
- (5) Remove the 4 screws (b). (See Fig.12)

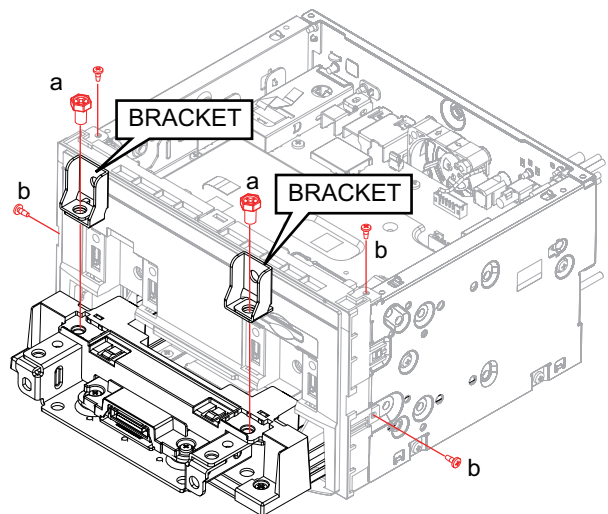


Fig.12

- (6) Remove the 6 hooks, and remove the SUB PANEL ASS'Y. (See Fig.13)

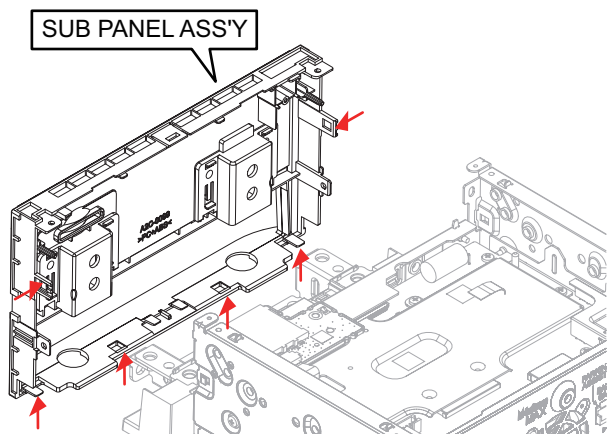


Fig.13

- (7) Disconnect the FFC from connector.
- (8) Remove the 2 screws, and remove the SD PWB. (See Fig.14)

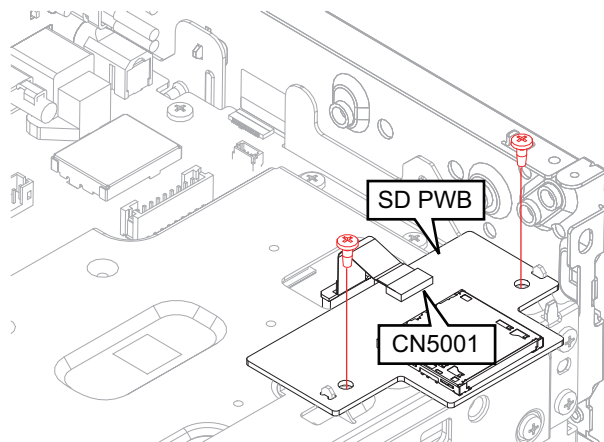


Fig.14

3.1.6 Removing the DIGITAL IO PWB (See Fig.15 to 18)

- (9) Disconnect the wire harness from connectors.
- (10) Remove the 4 screws, and remove the REAR PANEL. (See Fig.14)

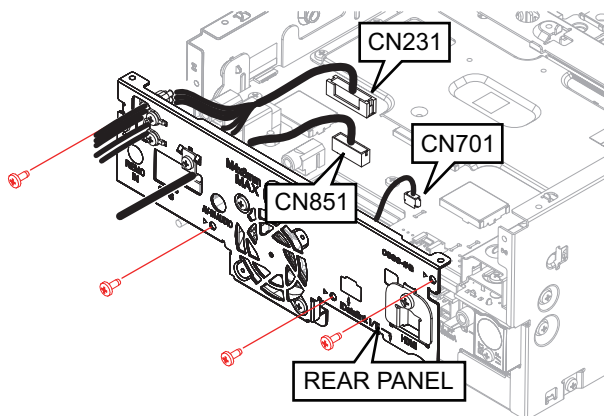


Fig.15

- (11) Remove the 8 screws, and remove the BRACKET. (See Fig.16)

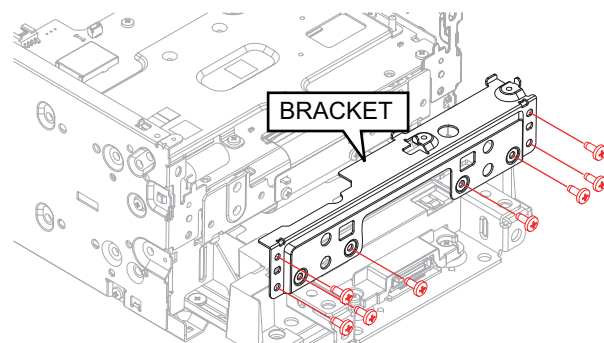


Fig.16

- (12) Remove the 10 screws, and remove the SIDE PANEL of the both side. (See Fig.17)

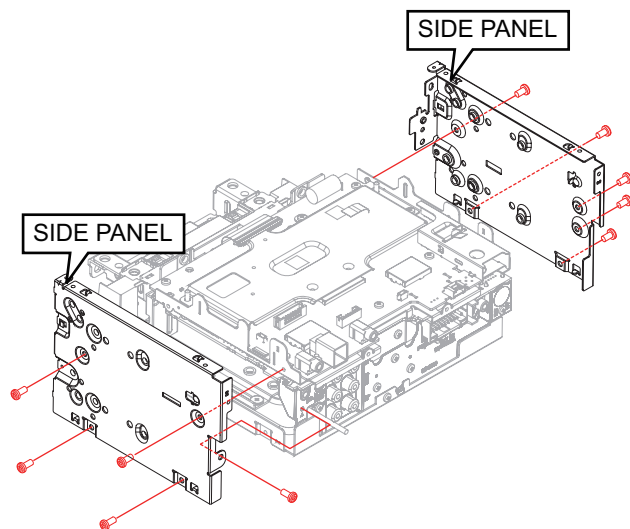


Fig.17

- (13) Disconnect the FPC from connector.
 (14) Remove the 4 screws (a), and remove the HEAT SINK. (See Fig.18)
 (15) Remove the 5 screws (b), and remove the DIGITAL IO PWB. (See Fig.18)

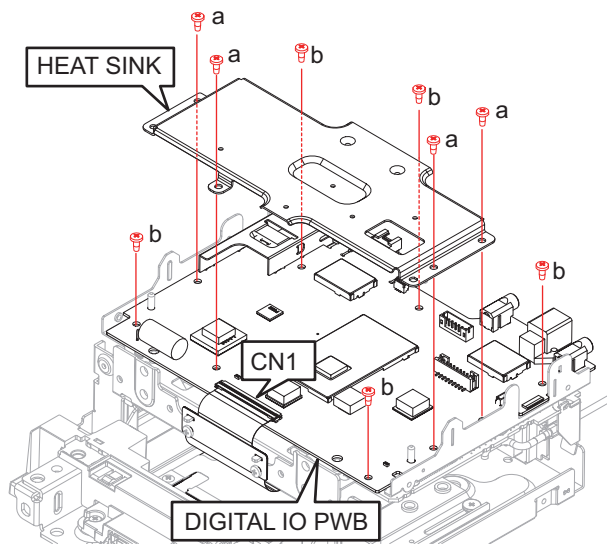


Fig.18

3.1.7 Removing the DETACH PWB (See Fig.19)

- (1) Remove the 2 screws (a), and remove the COVER.
 (2) Remove the 4 screws (b), and remove the COVER.
 (3) Disconnect the FPC from connector.
 (4) Remove the DETACH PWB.

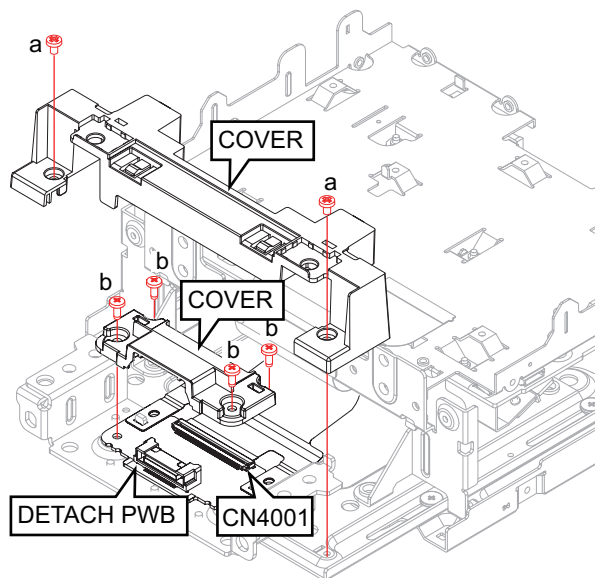


Fig.19

3.1.8 Removing the GYRO PWB (See Fig.20 and 21)

- (1) Remove the 3 screws, and remove the SUB CHASSIS ASS'Y. (See Fig.20)

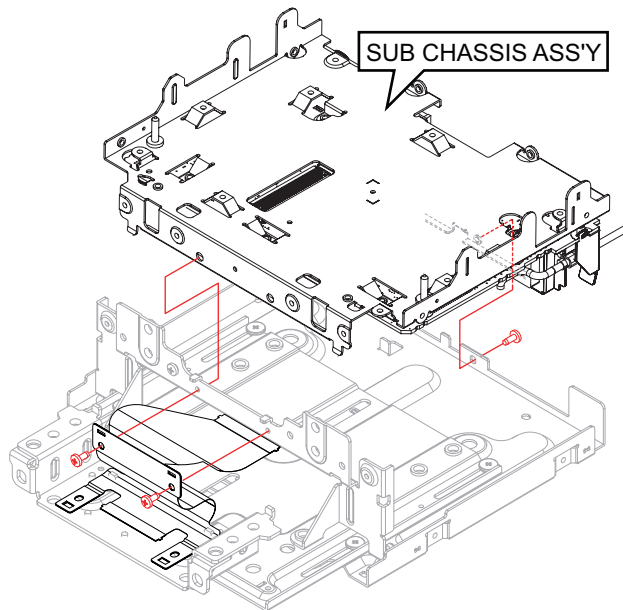


Fig.20

- (2) Disconnect the FFC from connector. (See Fig.21)
- (3) Remove the 2 screws, and remove the GYRO PWB.
(See Fig.21)

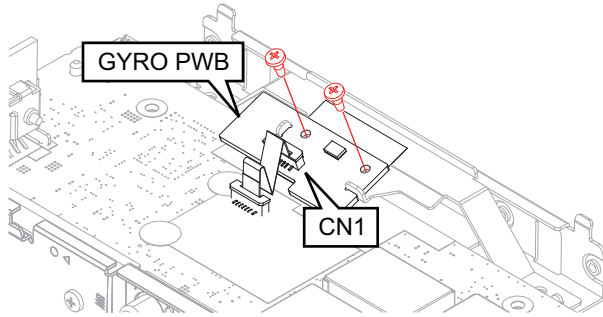


Fig.21

3.1.9 Removing the AUDIO PWB (See Fig.22)

- (1) Remove the 5 screws, and remove the AUDIO PWB.

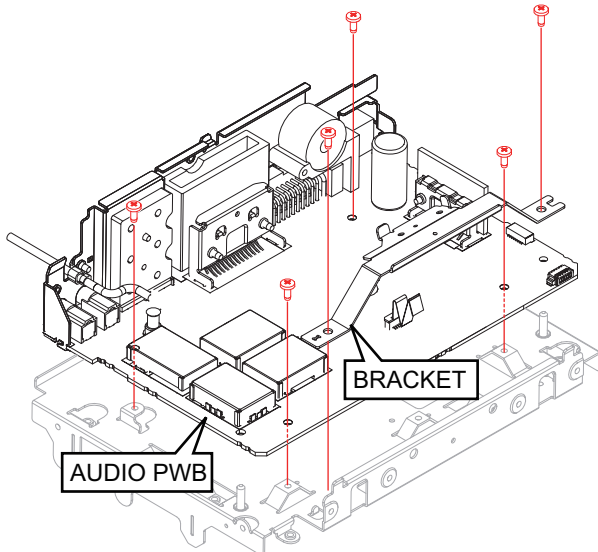
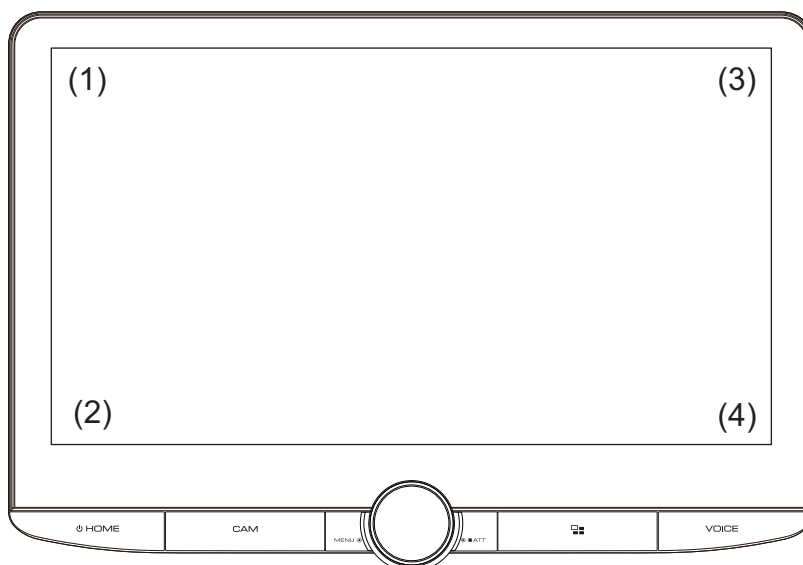


Fig.22

SECTION 4 ADJUSTMENT

4.1 Special Mode and Special operation List

Mode Name	Description	Prerequisite	Operation
TUNER span switching	For switching the frequency step / span of TUNER.	While displaying the STANDBY(AV OFF) screen.	Touch the screen within 2 seconds in the following order. Upper left corner (1) → Upper left corner (1) → Upper left corner (1) → Lower left corner (2) → Lower left corner (2) → Lower left corner (2)
SXM Code Clear (SXM compatible only)	For initializing the Parental Code of SiriusXM.		Touch "NVM Clear Passcode" after holding down the transparent button on the Direct Channel screen for 5 seconds.
Production Menu	Special function menu for the production line. (The production mode is turned on when the menu is displayed.)	While displaying the SETUP>System>Software Information screen.	Touch the screen within 2 seconds in the following order. Upper left corner (1) → Lower right corner (4) → Lower right corner (4) → Lower right corner (4) → Upper left corner (1) → Upper left corner (1)
Customize Menu (Opening screen / wallpaper / initial settings / steering remote control set rewrite mode)	For rewriting the configurations of the Opening screen / Wallpaper / Parameter / Learning Remote via USB.	While displaying the STANDBY(AV OFF) screen.	Touch the screen within 2 seconds in the following order. Upper left corner (1) → Upper left corner (1) → Lower left corner (2) → Lower left corner (2) → Upper left corner (1) → Lower left corner (2)
Service Menu	Special function menu used by servicing personnel.	While displaying the SETUP>System>Software Information screen.	Touch the screen within 2 seconds in the following order. Upper left corner (1) → Upper left corner (1) → Lower right corner (4) → Lower right corner (4) → Lower right corner (4) → Upper left corner (1)
Initialize Factory Default	For initializing the setting values / data to factory default.		Operation in the Production Menu/Service Menu/Developer Menu
Temporary Security Release (Security compatible model only)	For releasing the security temporarily when the security code is not known. Factory Initialize is required to clear the security.	While displaying the Security Code input screen.	Touch the screen within 2 seconds in the following order. Upper right corner (3) → Upper left corner (1) → Upper left corner (1) → Upper right corner (3) → Upper right corner (3) → Upper left corner (1)



4.2 TUNER span switching

Switching frequency initial value of the Tuner, band and frequency range, frequency step, preset frequency initial value, seek threshold value and the De-Emphasis value.

(1) In or STANDBY (KWD models) or AV OFF (JVC model) screen, touch the screen within 2 seconds in the following order.

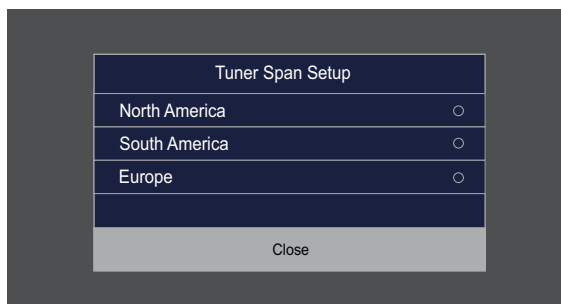
Upper left corner (1) → Upper left corner (1) → Upper left corner (1) → Lower left corner (2) → Lower left corner (2) → Lower left corner (2)



NOTE:

The screen display name (STANDBY or AV OFF) varies depending on the model.

(2) When the dialog is displayed, press the button according to the purpose. Flick up and down to scroll the screen.



NOTE:

Some items are not displayed by the model.

Key	Operation
North America	Set to the North America.
South America	Set to the South America.
Europe	Set to the Europe.
Asia General	Set to the Others.
Asia/Middle East	Set to the Middle East.
Australia	Set to the Australia.
Close	Closes this dialog box.

Setting value for each area

Item		North America	South America	Europe	Asia General	Asia/Middle East	Australia
frequency initial value (Hz)	LW/MW	1000k	1000k	999k	999k	999k	999k
	FM1	97.9M	97.9M	97.9M	97.9M	97.9M	97.9M
	FM2/FM3	87.9M	87.5M	87.5M	87.5M	87.5M	87.5M
frequency step (Hz)	LW/MW	10k	10k	9k	9k	9k	9k
	FM	200k	50k	50k	50k	50k	50k
Minimum frequency (Hz)	LW/MW	530k	530k	531k	531k	531k	531k
	FM	87.9M	87.5M	87.5M	87.5M	87.5M	87.5M
Maximum frequency (Hz)	LW/MW	1700k	1700k	1611k	1611k	1602k	1701k
	FM	107.9M	108.0M	108.0M	108.0M	108.0M	108.0M
Preset frequency initial value	FM1(Hz)	P1	87.9M	87.5M	87.5M	87.5M	87.5M
		P2	89.9M	89.9M	89.9M	89.9M	89.9M
		P3	97.9M	97.9M	97.9M	97.9M	97.9M
		P4	105.9M	105.9M	105.9M	105.9M	105.9M
		P5	107.9M	108.0M	108.0M	108.0M	108.0M
		P6	-	-	87.5M	87.5M	87.5M
		P6-P15	87.9M	87.5M	87.5M	-	-
	FM2/3 (Hz)	P1-P6	-	-	-	87.5M	87.5M
	AM (Hz)	P1	530k	530k	531k	531k	531k
		P2	600k	600k	603k	603k	603k
		P3	1000k	1000k	999k	999k	999k
		P4	1500k	1500k	1404k	1404k	1404k
		P5	1700k	1700k	1602k	1602k	1602k
		P6	-	-	-	531k	531k
Seek threshold value	LW/MW	30dBuVemf	30dBuVemf	30dBuVemf	30dBuVemf	30dBuVemf	30dBuVemf
	FM	15dBuV	15dBuV	17dBuV	12dBuV	12dBuV	12dBuV
De-emphasis value (us)	FM	75	75	50	50	50	50

*North America/South America/Europe presets are FM: P1 to P15, AM: P1 to P5.

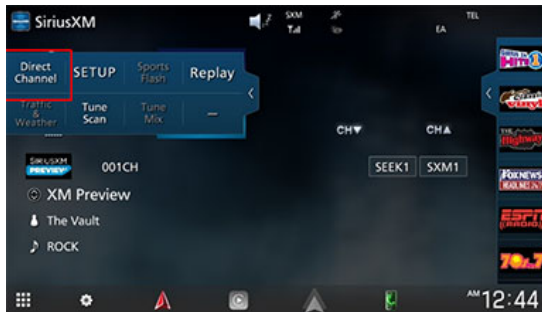
*Asia General, Asia/Middle East, Australia presets are P1 to P6 for each band.

*LW: 153 to 279kHz

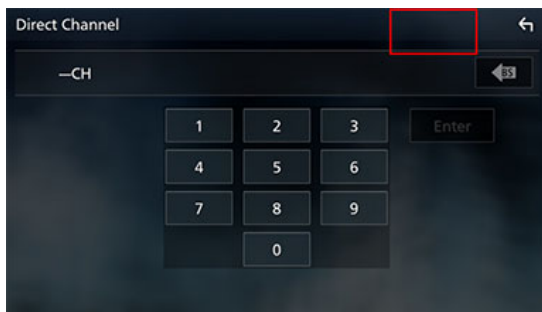
4.3 SiriusXM Code Clear Mode (SiriusXM compatible model only)

Initialize the Parental Code of SiriusXM.

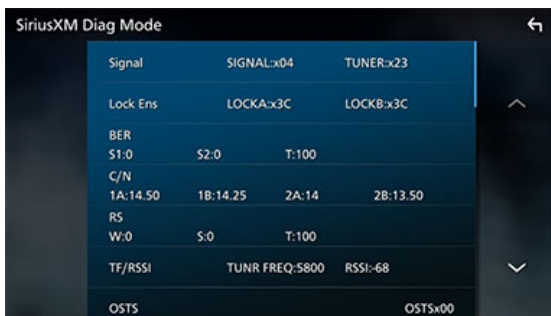
- (1) Set the source to SiriusXM.
- (2) Pull the left drawer, touch the "Direct Channel"



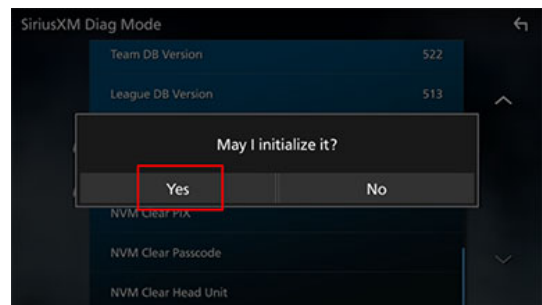
- (3) Touch the red mark of the part of the Direct Channel screen for 5 seconds.



- (4) Scroll to the bottom by dragging the list, select the "NVM Clear Passcode"

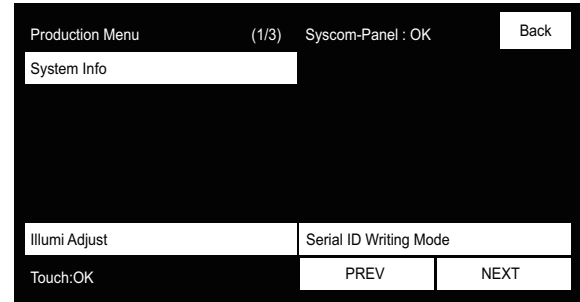


- (5) Touch the "Yes" to clear Passcode. (default value is 0000)



4.4 Production Menu Specification

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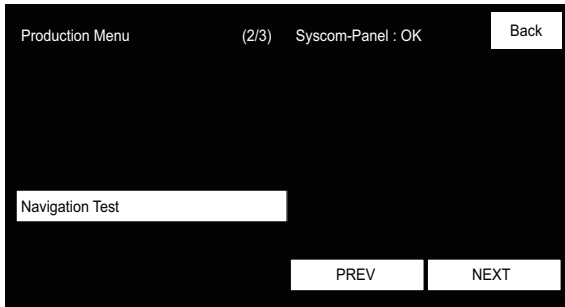
NOTE:

Some items are not displayed by the model.

Item	Description
Syscom-Panel	Displays the result (OK or NG) of the check on the communication between Syscom and the Panel CPU. *Test starts automatically when Production Menu starts.
Touch	Display the operation check result (OK or NG) of the electrostatic touch panel. Up to three error numbers are generated at the same time in the test1/2/3. Test1: E1 (Short of Y line and power supply) E2 (Short of Y line and GND) E3 (Short of Y line and Y line or failure of IC) E4 (Short of X line or failure of IC) Test2: E5 (Either X line or Y line is open) Test3: E6 (Inspection error) In case of simultaneous occurrence, it is displayed in one line delimited by "/". Example: E1 / E5 / E6

Key	Operation
System Info	Transition to the System Info screen.
Illumi Adjust	Transition to the Illumi Adjust screen. (Displayed in Illumi adjustment supporting model only) Button color is changed by the results. (Not performed: Black/OK: Blue/NG: Red)
Serial ID Writing Mode	This mode is not used for service.
NEXT	Switch the page from 1/3 to 2/3.
PREV	Switch the page from 1/3 to 3/3.
Back	Exit Production menu, and return to the normal screen

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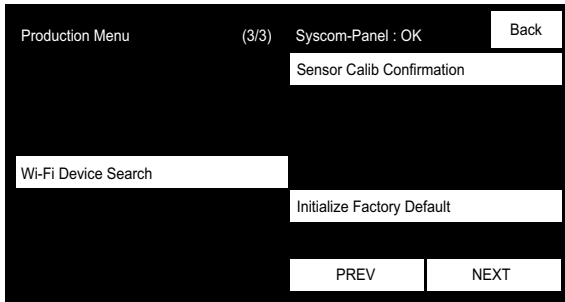
NOTE:

Some items are not displayed by the model.

Item	Description
Syscom-Panel	Displays the result (OK or NG) of the check on the communication between Syscom and the Panel CPU. *Test starts automatically when Production Menu starts.

Key	Operation
Navigation Test	This mode is not used for service. (Displayed in DNR model only)
NEXT	Switch the page from 2/3 to 3/3.
PREV	Switch the page from 2/3 to 1/3.
Back	Exit Production menu, and return to the normal screen

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NOTE:

Some items are not displayed by the model.

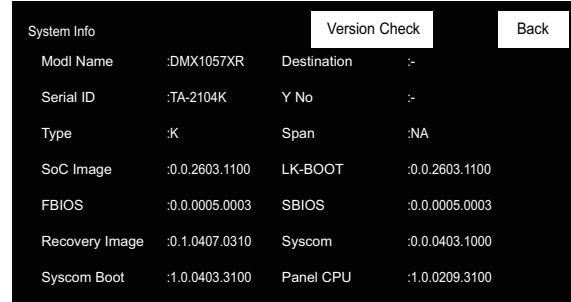
Item	Description
Syscom-Panel	Displays the result (OK or NG) of the check on the communication between Syscom and the Panel CPU. *Test starts automatically when Production Menu starts.

Key	Operation
Wi-Fi Device Search	This mode is not used for service.
Sensor Calib Confirmation	Transition to the Sensor Calib Confirmation screen.
Initialize Factory Default	Transition to the Initialize Factory Default screen.

4.4.1 SystemInfo (For Production Menu)

Mode for displaying the various versions of the system.

Click up and down to scroll the screen.



Item	Description
Model Name	Model name
Destination	"-" Fixed
Serial ID	Serial number Written condition: ***** (Alphanumeric) Not Written: -
Y No	Model number
Type	Type information K: North America R: Latin America E: Europe Q: Russia M: Asia (KWD: including Indonesia) M2: Middle East (JVC: including Indonesia) X: Australia A9: Indonesia
Span	Displaying Tuner span type (Tuner area) NA: North America EU: Europe SA: Latin America JP: Japan ASIA: Others BR: Brazil TN: Middle East AUS: Australia
SoC Image	SoC firmware version
BOOT	SoC BOOT version
FBIOS	SoC FBIOS version
SBIOS	SoC SBIOS version
Recovery Image	SoC recovery image version
Syscom	Syscom firmware version
Syscom Boot	Syscom boot version
Panel CPU	Panel CPU firmware version
Panel Unit	LCD manufacturer name
Touch Cnt	Touch Controller (ATMEL) version
eMMC	eMMC firmware version
Disc Unit Name	Disc mecha module name
Disc Version	Disc mecha module firmware version
Disc Region	DVD Region code
Disc CPRM	Disc mecha CPRM check sum

Item	Description
Disc Serial No	Disc mecha module serial no
Macrovision (Sys)	Macrovision version (System)
Macrovision (Disc)	Macrovision version (Disc mecha module)
HDCP Key	HDCP Key Condition (-/OK)
Navigation S/W	Navigation software Version *Displayed in DNR model ONLY
Base Map	Navigation database version (Base Map) *Displayed in DNR model ONLY
Detail Map	Navigation database version (Detail Map) *Displayed in DNR model ONLY
Junction View	Navigation database version (Junction View) *Displayed in DNR model ONLY
Dealer POI	Dealer POI Version *Displayed in OEM model only
DATA1	Opening data version
DATA2	Wallpaper version
DATA3	Steering Remote threshold version

After the version check is executed, items that are inconsistent with the version check are displayed in red.

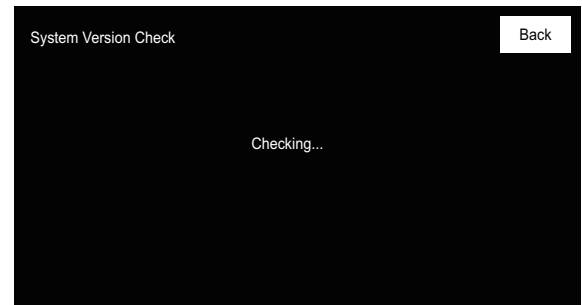
Key	Operation
Version Check	Perform the version check
Back	Return to the Production menu.

4.4.2 Version Check

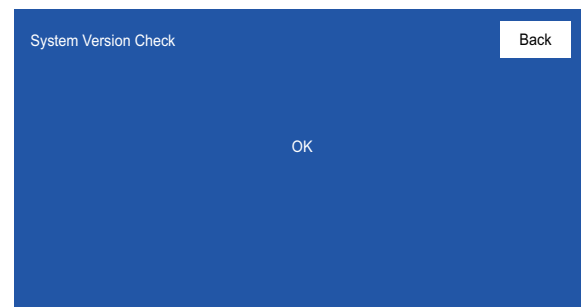
Mode for checking the version of each software/data.

- (1) Insert the USB memory that contains the version check data.
- (2) Select the "System Info" in the Production Menu, and displaying the System Info screen.
- (3) Press "Version Check" to start the version check and display the result.

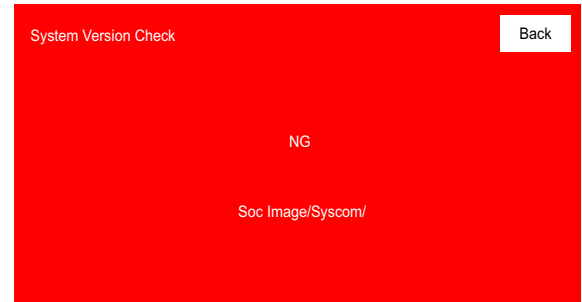
Checking...



Result OK



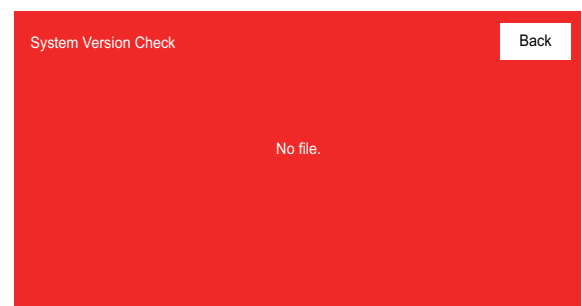
Result NG



This is displayed when the version information within the system is inconsistent with the version information of the version check data in the USB memory.

An item that is inconsistent is displayed below "NG" and the version value of the corresponding item in the System Info screen is displayed in red.

Result NG (no version check data)



This is displayed when there is no version check data in the USB memory.

Key	Operation
Back	Return to the Production menu.

4.4.2.1 For version check data

Store the following data in the root folder of the USB memory.

- File name: VCheck_model_system.txt
- File format:
Version Check
[(item name 1)](version name 1)
[(item name 2)](version name 2)
:
[(item name n)](version name n)
~EN

item name list

Item	Description
Model Name	Model name
Serial ID	Serial number
Yno	Model number
Type	Type information
Span	Tuner type information
SoC Image	SoC firmware version
LKBOOT	SoC BOOT version
FBIOS	SoC BIOS version
SBIOS	SoC SBIOS version
Recovery Image	SoC recovery image version
Syscom	Syscom firmware version
Syscom Boot	Syscom boot version
Panel CPU	Panel CPU firmware version
Panel Unit	LCD manufacturer name INNOLUX LCD: "INNOLUX" CPT LCD: "CPT" or "-" AUO LCD: "AUO" HANNSTAR LCD: "Hannstar"
Touch Cnt	Touch Controller (ATMEL) version
eMMC	FBIF version
Disc Unit Name	Disc mecha module name
Disc Version	Disc mecha module firmware version
Disc Region	DVD Region code
Disc CPRM	Disc mecha CPRM check sum
Disc Serial No	Disc mecha module serial no
Macrovision(Sys)	Macrovision(Sys)
Macrovision(Disc)	Macrovision(Disc)
HDCP Key	HDCP Key Condition (-/OK)
Navigation S/W	Navigation software version
Base Map	Navigation database version (Base Map)
Base Map Dscrpt	Navigation database description (Base Map)
Detail Map	Navigation database version (Detail Map)
Junction View	Navigation database version (Junction View)
Dealer POI	Dealer POI Version *OEM model only
DATA1	Opening data version
DATA2	Wallpaper version
DATA3	Steering Remote threshold version

- Checks on items not included in the version check data are skipped.
- If the version name is "-", the absence of the corresponding data is checked.
- If the version name is "!", the presence of the corresponding data is checked.

4.4.2.2 Adjustment / test flag check

The following items are checked (OK/NG) during the version check. Items that are not applicable are not checked.

- LCD Adjust
- Panel Mecha Adjust
- GPS Test
- Serial ID
- key illumi adjustment
- Region Code writing
- Touch panel adjustment
- HD-Radio Test
- SXM Test
- DAB Test
- iDataLink Test
- Tuner Smeter Test
- BT Device Search
- Gesture Test
- Touch Panel Test
- iPod Test
- Learning Remote Test
- SD Card R/W Test @ SRS-92-3126
- HDMI/CEC Test @SRS-92-3141
- Unit ID API Check @SRS-92-3216
- TMC Tuner @SRS-92-3214
- Ext NaviBox Test @SRS-92-3143

4.4.3 Illumi Adjust (Key Illumi supporting model only)

Mode for adjusting the white of Key illumi.

- (1) Select the "Illumi Adjust" in the Production Menu.
- (2) Make sure that as much as more "White color" key illumination condition by 10 kinds of option.
- (3) Press the "OK" to save the value.

Illumination Adjust		Back
pw05, pw09	pw10	
pw12	pw01	
pw11	pw08	
pw17	pw02, pw06, pw07	
pw04, pw20	pw03, pw18, pw19	
		OK

Save OK

Illumination Adjust		Back
pw05, pw09	pw10	
pw12	pw01	
pw11	pw08	
pw17	pw02, pw06, pw07	
pw04, pw20	pw03, pw18, pw19	
- SAVE Success -		OK

Save NG

Illumination Adjust		Back
pw05, pw09	pw10	
pw12	pw01	
pw11	pw08	
pw17	pw02, pw06, pw07	
pw04, pw20	pw03, pw18, pw19	
- SAVE Error -		OK

Key	Operation
(pw05)	Switch the LED duty (R=72, G=80, B=48)
(pw13)	Switch the LED duty (R=81, G=62, B=62)
(pw11, pw12)	Switch the LED duty (R=55, G=74, B=58)
(pw09, pw10)	Switch the LED duty (R=63, G=69, B=48)
(pw14, pw15)	Switch the LED duty (R=63, G=71, B=58)
(pw08)	Switch the LED duty (R=48, G=80, B=59)
(pw16)	Switch the LED duty (R=53, G=62, B=71)
(pw06, pw07)	Switch the LED duty (R=53, G=69, B=48)
(pw03, pw04)	Switch the LED duty (R=48, G=88, B=57)
(pw01, pw02)	Switch the LED duty (R=70, G=80, B=43)
Back	Return to the Production menu.

The font color of the "Illumi Adjust" button on the first page of the Production Menu changes depending on the result.

- Not performed: Black
- OK: Blue
- NG: Red

4.4.4 Sensor Calib Confirmation

Mode for displaying the writing result of 6-axis sensor correction value

Sensor Calib Confirmation	Back
RESULT[1] GyroOfs[32810.612500, 32823.387500, 32781.900000] GyroTemp[152.000000, 152.000000, 152.000000] GyroSCW[0.003792655522580526, 0.003839250578287114, 0.003813312910924187] GyroValid[1, 1, 1] AccOfs[2062.612500, 2035.087500, 2031.312500] AccTemp[136.412500, 136.412500, 138.562500] AccSCW[0.009645746884778291, 0.009615931518323547, 0.009587339606501282] AccValid[1, 1, 1] ProdPlt[JKNG] Serial ID[XJS005K00A040A70010000] sYSuNITId[XXXXXXXXXX]	

Item	Description
RESULT	Display the writing result of 6-axis sensor correction value. 00H: NG/01H: OK
GyroOfs	Gyro sensor X-axis/Y-axis/Z-axis offset (state of rest) (IEEE 754 double precision floating point number)
GyroTemp	Gyro sensor average temperature when calculating X-axis/Y-axis/Z-axis offset (IEEE 754 double precision floating point number)
GyroSCW	Gyro sensor X-axis/Y-axis/Z-axis sensitivity coefficient calculation value (clockwise)

Item	Description
GyroValid	Enable or disable for gyro sensor X axis/Y axis/Z axis 00H: Enable/01H: Disable
AcclOfs	Accelerometer X-axis / Y-axis / Z-axis offset (state of rest X-axis / Y-axis / Z-axis are horizontal)
AcclTemp	Acceleration sensor average temperature when calculating X-axis/Y-axis/Z-axis offset (IEEE 754 double precision floating point number)
AcclSCW	Acceleration sensor X-axis/Y-axis/Z-axis sensitivity coefficient calculation value (clockwise)
AcclValid	Enable or disable for acceleration sensor X axis/Y axis/Z axis 00H: Enable/01H: Disable
ProdPlt	Production area (9 bytes ASCII character string including NULL terminal character)
Serial ID	Board serial number (32 bytes 22 bytes ASCII character string and Fill the rest with null terminal character)
SysUnilD	Syscon Uniqu ID 128bit

Key	Operation
Back	Return to the Production menu.

4.4.5 Initialize Factory Default (for Production Menu)

Mode for initializing to factory default.

Initialize all of the backup memory of the system.

The flag for changing the font color of the button according to the OK/NG status in the Production Menu is also cleared.

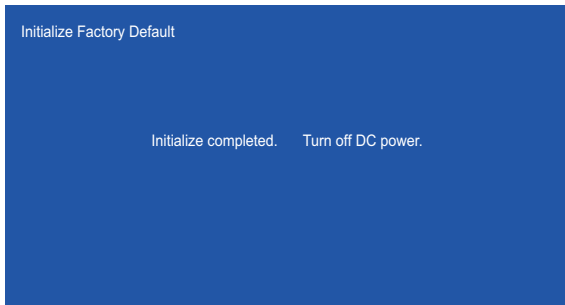
The following data are not initialized.

- LCD adjustment value
 - Panel mecha adjustment value
 - Key illumi adjustment value
 - Gesture sensor adjustment value
 - Model number / serial ID
 - Region code
 - Tuner Area Setting
 - Opening customize data
 - Wallpaper customize data
 - Initial value customize data
 - Learning remote control customize data
- (1) Select the "Initialize Factory Default" in the Production Menu.
 - (2) A confirmation dialog box appears, select "Yes" to starts the factory initialize.

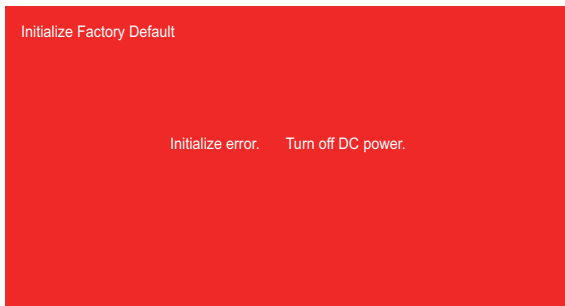
Initializing...



Processing is completed (Normal end)



Processing is completed (When the error is occurred.)



This unit will not be restarted after the initialization process is complete, turn off/on the power to confirm as necessary.

4.5 Customize Menu Specification

Customize Menu		Back
Opening Customize	Initialize Factory Default	
Wallpaper Customize		
Parameter Customize	R-CAM Guideline Parameter	
Learning Remote Customize		
Delete Base Map		
Restore Base Map		

NOTE:

Some items are not displayed by the model.

Key	Operation
Opening Customize	Transition to the Opening Customize Screen.
Wallpaper Customize	Transition to the Wallpaper Customize Screen.
Parameter Customize	Transition to the Parameter Customize Screen.
Learning Remote Customize	Transition to the Learning Remote Customize Screen.
Delete Base Map	Transition to the Delete Base Map Screen.
Restore Base Map	Transition to the Restore Base Map Screen.
Initialize Factory Default	Transition to the Initialize Factory Default Screen.
R-CAM Guideline Parameter	Transition to the R-CAM Guideline Parameter Screen.
Back	Exit the customize menu, return to the normal screen.

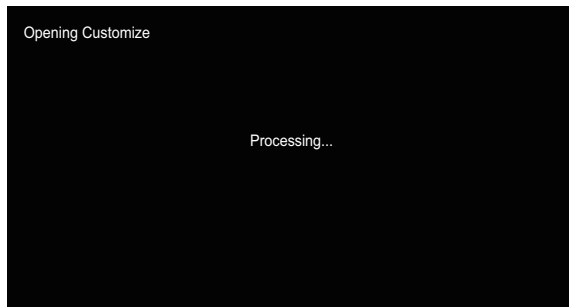
4.5.1 Opening Customize

Mode for customize of startup screen.

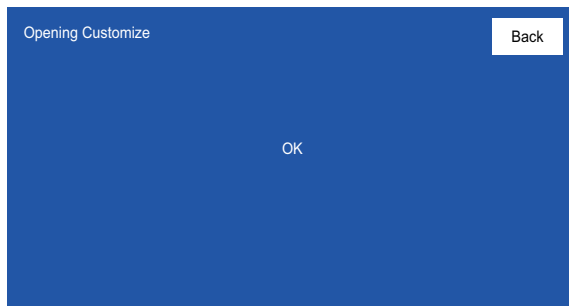
When restoring to default

- If the version number is 0000, the opening image is restored to the consumer default image.
 - Download the region kit file from Service Web page. The file name of the kit is "Dummy_File*.zip".
 - Extract this file to your computer desktop.
 - Choose appropriate region files from the above folder.
 - Open folder and burn "sample_wallpaper.bmp" to root of USB MEMORY.
- (1) Connect the USB memory containing the Opening Customize data.
 - (2) When "Opening Customize" is selected in the Production Menu or Customize Menu, the customize process starts and the result is displayed.

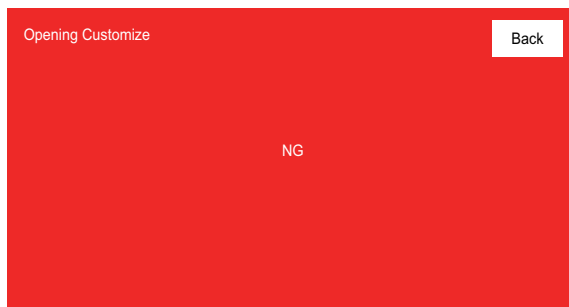
Checking...



Customize OK



Customize NG



For opening customize data

Store the following data in the root folder of the USB memory.

- Setting information file
 - File name: OpeningCustomize.txt
 - File format:
Opening Customize
[FILE] (File name)
[VERSION] (Version number 4-digit number)
~END

*File name is opening image file name.

If the version number is 0000, the opening image is restored to the consumer default image.

- Image file
 - Format: 16bit RGB565 bitmap
 - Image size: 800x480
 - File name: optional
- *Image content for LCD with an aspect ratio of one pixel.

<Entry example of the setting information file>

Opening Customize
[FILE]sample_opening.bmp
[VERSION]9876
~END

4.5.2 Wallpaper Customize

Mode for customize of wallpaper.

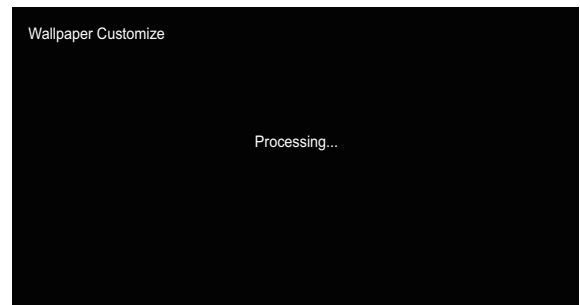
When restoring to default

- If the version number is 0000, the opening image is restored to the consumer default image.
- Download the region kit file from Service Web page. The file name of the kit is "Dummy_File*.zip".
- Extract this file to your computer desktop.
- Choose appropriate region files from the above folder.
- Open folder and burn "sample_wallpaper.bmp" to root of USB MEMORY.

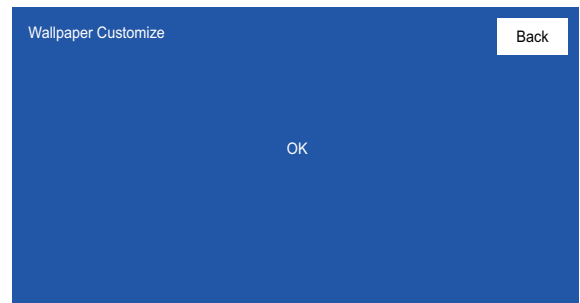
(1) Connect the USB memory containing the wallpaper customize data.

(2) When "Wallpaper Customize" is selected in the Production Menu or Customize Menu, the customize process starts and the result is displayed.

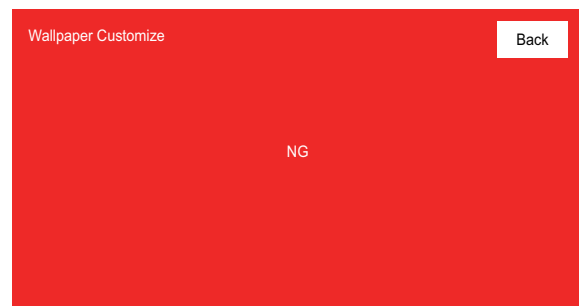
Checking...



Customize OK



Customize NG



For wallpaper customization data

Store the following data in the root folder of the USB memory.

- Setting information file
 - File name: WallpaperCustomize.txt
 - File format:
Wallpaper Customize
[FILE] (File name)
[VERSION] (Version number 4-digit number)
~END

*File name is opening image file name.

If the version number is 0000, the opening image is restored to the consumer default image.

- Image file

- Format: 16bit RGB565 bitmap
- Image size: 800x480
- File name: optional

*Image content for LCD with an aspect ratio of one pixel.

<Entry example of the setting information file>

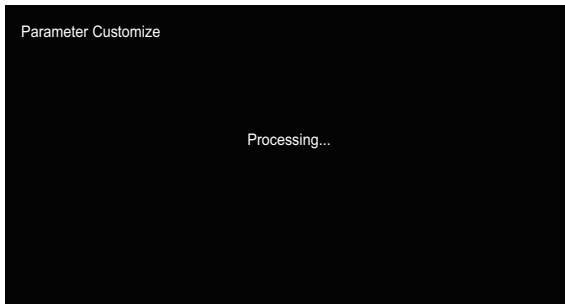
```
Wallpaper Customize
[FILE]sample_wallpaper.bmp
[VERSION]9876
~END
```

4.5.3 Parameter Customize

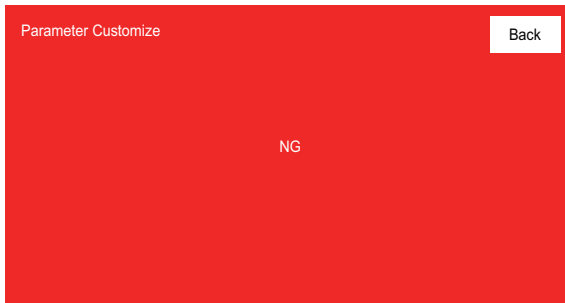
Mode for customize of initial value.

- (1) Connect the USB memory containing the initial value customize data.
- (2) When "Parameter Customize" is selected in the Production Menu or Customize Menu, the customize process starts and the result is displayed.

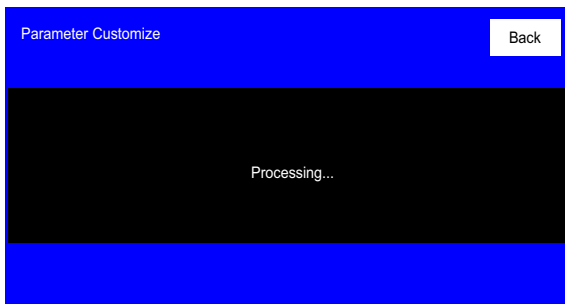
Checking...



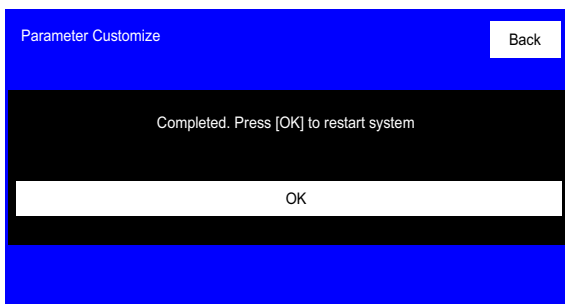
Customize NG



Customize OK → Initializing factory default in progress



Initializing factory default completed → Restart by pressing OK button (Complete after restart)



For initial value customize data

Store the following data in the root folder of the USB memory.

- Setting information file
 - File name: DOP_Customize.txt
 - File format: Refer to the following table.

Line No	Setting items	Settings
1	DOP_Customize	Header (fixed character string)
2	[model]	Model setting
3	[maker]	Maker setting (fixed to "2")
4	[illumi_1]	Panel color settings (when the Dimmer Off) 0:RED 1:PURPLE 2:PINK 3:AMBER 4:GREEN 5:AQUA *Not supported 6:BLUE 7:WHITE *Not supported 8:SKYBLUE 9:YELLOW 10:SCAN
5	[illmiuser_Red]	Panel color settings (when the Dimmer Off) when the user setting Red value
6	[illmiuser_Green]	Panel color settings (when the Dimmer Off) when the user setting Green value
7	[illmiuser_Blue]	Panel color settings (when the Dimmer Off) when the user setting Blue value

Line No	Setting items	Settings
8	[lang]	language 0: English (United States) 1: English (UK) 2: Dutch 3: French 4: German 5: Italian 6: Spanish 7: Arabic (World Arabic) 8: Persian 9: Hebrew A: Bulgarian *Not supported B: Ukrainian * Not supported C: Swedish *Not supported D: Slovenian *Not supported E: Slovak *Not supported F: Serbian *Not supported 10: Romanian *Not supported 11: Norwegian *Not supported 12: Lithuanian *Not supported 13: Latvian *Not supported 14: Finnish *Not supported 15: Danish *Not supported 16: Croatian *Not supported 17: Taiwanese *DNX/DDX/DMX not supported 18: Chinese 19: Estonian *Not supported 1A: Greek *KW-V/KW-M not supported 1B: Malay (Malaysia) 1C: Indonesian 1D: Portuguese 1E: Russian 1F: Thai 20: Turkish 21: Czech *KW-V/KW-M not supported 22: Polish *KW-V/KW-M not supported 23: Hungarian *KW-V/KW-M not supported 24: Japanese *Not supported 25: Korean *Not supported 26: English (Australia) *Not supported 27: Arabic (Saoudian) *Not supported 28: Arabic (Gulf Arabic) *Not supported 29: Catalunya *Not supported 2A: Flemish *Not supported 2B: French (Canada) *Not supported 2C: Portuguese (Brazil) *Not supported 2D: Spanish (Mexico) *Not supported
9	[demo]	Demo mode setting 0:Off 1:On
10	[rearcam]	Rear camera interrupt setting 0:Off 1:On
11	[pguideline]	Rear camera guide line display 0:Off 1:On
12	[glinevga_ulx]	Rear camera guide line setting (back left) * VGA model X coordinate 08h: 8 to 319h:793

Line No	Setting items	Settings
13	[glinevga_uly]	Rear camera guide line setting (back left) * VGA model Y coordinate 0Ah: 10 to 190h: 400
14	[glinevga_urx]	Rear camera guide line setting VGA (back right) * VGA model X coordinate 08h: 8 to 319h: 793
15	[glinevga_ury]	Rear camera guide line setting VGA (back right) * VGA model Y coordinate 0Ah: 10 to 190h: 400
16	[glinevga_llx]	Rear camera guide line setting VGA (front left) * VGA model X coordinate 08h: 8 to 319h: 793
17	[glinevga_lly]	Rear camera guide line setting VGA (front left) * VGA model Y coordinate 0Ah: 10 to 190h: 400
18	[glinevga_lrx]	Rear camera guide line setting VGA (right front) * VGA model X coordinate 08h: 8 to 319h: 793
19	[glinevga_lry]	Rear camera guide line setting VGA (right front) * VGA model Y coordinate 0Ah: 10 to 190h: 400
20	[glinevga_center]	Rear camera guide line setting VGA (middle guide line) * VGA model 00h: 0% 01h: 5% 02h: 10% 03h: 15% 04h: 20% 05h: 25% 06h: 30% 07h: 35% 08h: 40% 09h: 45% 0Ah: 50% 0Bh: 55% 0Ch: 60% 0Dh: 65% 0Eh: 70% 0Fh: 75% 10h: 80% 11h: 85% 12h: 90% 13h: 95% 14h: 100%
21	[glinehd_ulx]	Rear camera guide line setting (back left) * HD model X coordinate 0Dh: 13 to 4F5h:1269

Line No	Setting items	Settings
22	[glinehd_uly]	Rear camera guide line setting (back left) *HD model Y coordinate 0Fh: 15 to 258h: 600
23	[glinehd_urx]	Rear camera guide line setting HD (back right) *HD model X coordinate 0Dh: 13 to 4F5h: 1269
24	[glinehd_ury]	Rear camera guide line setting HD (back right) *HD model Y coordinate 0Fh: 15 to 258h: 600
25	[glinehd_llx]	Rear camera guide line setting HD (front left) *HD model X coordinate 0Dh: 13 to 4F5h: 1269
26	[glinehd_lly]	Rear camera guide line setting HD (front left) *HD model Y coordinate 0Fh: 15 to 258h: 600
27	[glinehd_lrx]	Rear camera guide line setting HD (right front) *HD model X coordinate 0Dh: 13 to 4F5h: 1269
28	[glinehd_lry]	Rear camera guide line setting HD (right front) *HD model Y coordinate 0Fh: 15 to 258h: 600
29	[glinehd_center]	Rear camera guide line setting HD (middle guide line) *HD model 00h: 0% 01h: 5% 02h: 10% 03h: 15% 04h: 20% 05h: 25% 06h: 30% 07h: 35% 08h: 40% 09h: 45% 0Ah: 50% 0Bh: 55% 0Ch: 60% 0Dh: 65% 0Eh: 70% 0Fh: 75% 10h: 80% 11h: 85% 12h: 90% 13h: 95% 14h: 100%

Line No	Setting items	Settings
30	[miclevel]	Bluetooth settings (microphone gain) 00h: -10 01h: -9 02h: -8 03h: -7 04h: -6 05h: -5 06h: -4 07h: -3 08h: -2 09h: -1 0Ah: 0 0Bh: 1 0Ch: 2 0Dh: 3 0Eh: 4 0Fh: 5 10h: 6 11h: 7 12h: 8 13h: 9 14h: 10
31	[echocancel]	Bluetooth setting (Echo cancel) 00h: -5 01h: -4 02h: -3 03h: -2 04h: -1 05h: 0 06h: 1 07h: 2 08h: 3 09h: 4 0Ah: 5
32	[noizereduction]	Bluetooth setting (Noise reduction) 00h: -5 01h: -4 02h: -3 03h: -2 04h: -1 05h: 0 06h: 1 07h: 2 08h: 3 09h: 4 0Ah: 5
33	[rcamera_message]	Rear camera warning message display setting 0: Always display 1: Clear after 5 seconds
34	[caution]	Caution screen rewriting at startup Png file name
35	~END	Termination marker (fixed character string)

*Please note that the languages supported differ depending on the models.

<Entry example of the setting information file>

```
DOP_Customize
[model]2
[maker]2
[illumi_1]A
[illumiuser_Red]1A
[illumiuser_Green]3
[illumiuser_Blue]C
[illumi_2]2
[lang]3
[demo]1
[rearcam]1
[pguideline]1
[glinevga_ulx]12C
[glinevga_uly]78
[glinevga_urx]1F4
[glinevga_ury]78
[glinevga_lx]64
[glinevga_ly]168
[glinevga_lrx]2BC
[glinevga_lry]168
[glinevga_center]B
[glinehd_ulx]12C
[glinehd_uly]C8
[glinehd_urx]1F4
[glinehd_ury]78
[glinehd_lx]44C
[glinehd_ly]168
[glinehd_lrx]421
[glinehd_lry]168
[glinehd_center]14
[miclevel]9
[echocancel]A
[noizereduction]3
[rcamera_message]1
[caution]Caution.png
~END
```

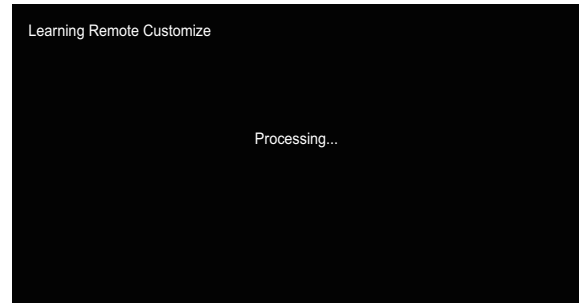
4.5.4 Learning Remote Customize

This function is used only steering remote control model.

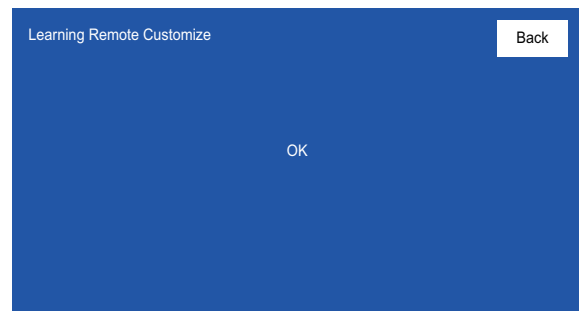
Mode for key assignment customize of learning remote control.

- (1) Connect the USB memory containing the remote control customize data.
- (2) When "Learning Remote Customize" is selected in the Production Menu or Customize Menu, the customize process starts and the result is displayed.

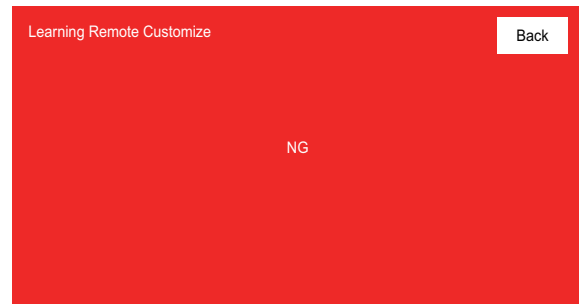
Checking...



Customize OK



Customize NG



For remote control customize data

Store the following data in the root folder of the USB memory.

- Setting information file
 - File name: SteRemoCustomize.txt
 - File format: Refer to the following table.

Line No	Setting items	Settings
1	Steering Remote Customize	Header (fixed character string)
2	[VERSION]	ST.Remo threshold value data version information (hexadecimal: 0002 to FFFF) * 1 * 2
3	[SET_FUNC]	Function permitted to user 0: SET/INITIALIZE 1: SET 2: NOTHING
4		Feed code

Line No	Setting items	Settings
5	[PULLUP]	Pull-up resistor value 0: 460Ω 1: 4.7kΩ
6	<OFF_1>[PRE]** **[MIN]****[MAX]* ***	OFF threshold value of the AD port 1 [PRE] Reference value * 3 [MIN] Lower limit value * 3 [MAX] Upper limit value * 3
7	<OFF_2>[PRE]** **[MIN]****[MAX]* ***	OFF threshold value of the AD port 2 *Meaning of the parameters are same as OFF threshold value of the AD port 1.
8		Feed code
9	<01>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.1 [FUNC] Function (see function table) [PRE] Reference value * 3 [MIN] Lower limit value * 3 [MAX] Upper limit value * 3 [PORT] Port 0: AD Port 1 1: AD port 2
10	<02>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.2 *Meaning of the parameters are same as ON threshold value of the key No.1.
11	<03>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.3 *Meaning of the parameters are same as ON threshold value of the key No.1.
12	<04>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.4 *Meaning of the parameters are same as ON threshold value of the key No.1.
13	<05>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.5 *Meaning of the parameters are same as ON threshold value of the key No.1.
14	<06>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.6 *Meaning of the parameters are same as ON threshold value of the key No.1.
15	<07>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.7 *Meaning of the parameters are same as ON threshold value of the key No.1.
16	<08>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.8 *Meaning of the parameters are same as ON threshold value of the key No.1.
17	<09>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.9 *Meaning of the parameters are same as ON threshold value of the key No.1.
18	<10>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.10 *Meaning of the parameters are same as ON threshold value of the key No.1.

Line No	Setting items	Settings
19	<11>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.11 *Meaning of the parameters are same as ON threshold value of the key No.1.
20	<12>[FUNC]**[P RE]****[MIN]****[MAX]****[PORT]*	ON threshold value of key No.12 *Meaning of the parameters are same as ON threshold value of the key No.1.
21	~END	Termination marker (fixed character string)

*1: If the version is set to 0001, the data written from the USB is cleared and the data in the unit is the same as the default model.

*2: 0000 is not written to the version information.

*3: Hex: 0000 ~ 03FF, Unregistered: FFFF

Function number	Function
0	VOL+
1	VOL-
2	ATT
3	SRC
4	⏮
5	⏭
6	ON HOOK
7	OFF HOOK
8	TEL
9	STANDBY
A	Radio/HD-Radio
B	DAB/Digital Radio
C	DISC *Not supported
D	AV-IN/TV
E	AV IN2
F	USB
10	iPod
11	Bluetooth
12	Mirror Link *Not supported
13	HDMI/MHL
14	Pandora *Not supported
15	Aupee Source *Not supported
16	Spotify *Not supported
17	Apple CarPlay/Android Auto/Mirroring
18	▶
19	FM+
1A	AM-
1B	OPEN
1C	NAV
1D	VOICE
1E	MUTE
1F	BAND
20	Preset Seek Down *Not supported
21	Preset Seek Up *Not supported
22	Angle *Not supported

Function number	Function
23	Disp *Not supported
24	SiriusXM *Not supported
25	SD
FE	No Function *Not supported
FF	No Function *Not supported

<Entry example of the setting information file>

Steering Remote Customize

[VERSION]0002

[SET_FUNC]0

[PULLUP]1

<OFF_1>[PRE]3FB[MIN]3D0[MAX]3FF

<OFF_2>[PRE]3FF[MIN]3D3[MAX]3FF

<01>[FUNC]1[PRE]60[MIN]34[MAX]8B[PORT]0

<02>[FUNC]2[PRE]C9[MIN]9E[MAX]F4[PORT]0

<03>[FUNC]4[PRE]142[MIN]117[MAX]16D[PORT]0

<04>[FUNC]5[PRE]1FC[MIN]1D1[MAX]227[PORT]0

<05>[FUNC]6[PRE]25F[MIN]234[MAX]28B[PORT]0

<06>[FUNC]14[PRE]2C9[MIN]29D[MAX]2F4[PORT]0

<07>[FUNC]FF[PRE]FFFF[MIN]FFFF[MAX]FFFF[PORT]0

<08>[FUNC]FF[PRE]FFFF[MIN]FFFF[MAX]FFFF[PORT]0

<09>[FUNC]FF[PRE]FFFF[MIN]FFFF[MAX]FFFF[PORT]0

<10>[FUNC]FF[PRE]FFFF[MIN]FFFF[MAX]FFFF[PORT]0

<11>[FUNC]FF[PRE]FFFF[MIN]FFFF[MAX]FFFF[PORT]0

<12>[FUNC]FF[PRE]FFFF[MIN]FFFF[MAX]FFFF[PORT]0

~END

4.5.5 Base Map delete (DNX model only)

- (1) Select the "Delete Base Map" in the Customize Menu.
- (2) A confirmation dialog box appears, select "Yes" to delete the base map.

Note: Do not press any hard keys on the unit while the process is ongoing.

4.5.6 Base Map restore (DNX model only)

- (1) Download the gmapbmap.zip file from CUSiS.
- (2) Unzip the file and copy the gmapbmap.img file to the USB root directory.
- (3) Insert the USB memory.
- (4) Select the "Restore Base Map" in the Customize Menu.
- (5) A confirmation dialog box appears, select "Yes" to restore the base map.

Note: Do not press any hard keys on the unit while the process is ongoing.

4.5.7 R-CAM Guideline Parameter

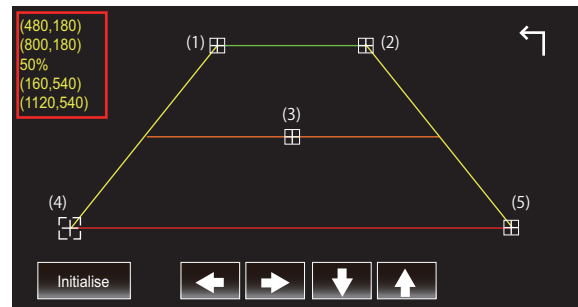
Mode for displaying the for rear camera guideline of the adjustment point coordinates.

- (1) Touch "R-CAM Guideline Parameter" in the Customize Menu.
- (2) The display changes to orange and the guideline adjustment display turns ON.
- (3) Select "Back" in the Customize Menu, and return to the normal screen.

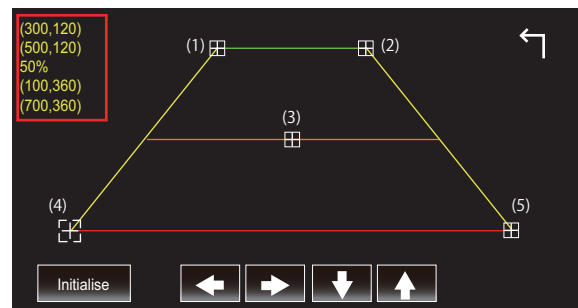
4.5.7.1 About display of guideline adjustment points

- (1) Transit to guideline adjustment screen of rear camera with SETUP> Camera> Guidelines SETUP.

WHD LCD



WVGA LCD



Define the upper left corner of the screen as the origin (0, 0), and display the coordinates of the guide line with the upper left point as (1), the upper right point as (2), the lower left point as (3), the lower right point as (4).

The straight line connecting the points obtained by dividing (1) to (4) and (2) to (5) into 20 is (3).

Display in 5% increments with assuming that the coordinates of (1) are 0% and the coordinates of (5) are 100%.

<Coordinate display example of WHD>

(1) (480,180)
 (2) (800,180)
 (3) 50%
 (4) (160,540)
 (5) (1120,540)

4.6 Service Menu Specification

Service Menu		Back
System Info	Sensor Status	
Error Log	Software Update	
Transfer Data	Clear Data	
Serial ID Manual Input	Y No Manual Input	
LCD Adjust	UnitID Transration	
NCM MAC Address		

NOTE:

Some items are not displayed by the model.

Key	Operation
System Info	Transition to the System Info screen.
Sensor Status	Transition to the Sensor Status screen.
Error Log	Transition to the Error log screen.
Software Update	Transition to the Software Update screen.
Transfer Data	Transition to the Transfer the data screen.
Clear Data	Transition to the Clear Data screen.
Serial ID Manual Input	Transition to the manual input of the serial ID screen.
Y No Manual Input	Transition to the Y No Manual Input screen. (Displayed in DNR model only)
LCD Adjust	Transition to the LCD Adjust screen.
UnitID Transration	Transition to the UnitID Translation screen. (Displayed in DNR model only)
NCM MAC Address	Transition to the NCM MAC Address screen.
RealVNC Menu	Transition to the RealVNC Menu screen.
Back	Exit Service menu, and return to the normal screen.

4.6.1 SystemInfo (For Service Menu)

Mode for displaying the various versions of the system.
Click up and down to scroll the screen.

System Info				Export	Back
Modl Name	:DMX1057XR	Destination	: -		
Serial ID	:TA-2104K	Y No	: -		
Type	:K	Span	:NA		
SoC Image	:0.0.2603.1100	LK-BOOT	:0.0.2603.1100		
FBIOS	:0.0.0005.0003	SBIOS	:0.0.0005.0003		
Recovery Image	:0.1.0407.0310	Syscom	:0.0.0403.1000		
Syscom Boot	:1.0.0403.3100	Panel CPU	:1.0.0209.3100		

Item	Description
Model Name	Model name
Destination	"-" Fixed
Serial ID	Serial number Written condition: ***** (Alphanumeric) Not Written: -
Y No	Model number
Type	Type information K: North America R: Latin America E: Europe Q: Russia M: Asia (KWD: including Indonesia) M2: Middle East (JVC: including Indonesia) X: Australia A9: Indonesia
Span	Tuner type information Displaying Tuner span type (Tuner area) NA: North America EU: Europe SA: Latin America JP: Japan ASIA: Others BR: Brazil TN: Middle East AUS: Australia ???: Unknown
SoC Image	SoC firmware version
LKBOOT	SoC LKBOOT version
FBIOS	SoC FBIOS version
SBIOS	SoC SBIOS version
Recovery Image	SoC recovery image version
Syscom	Syscom firmware version
Syscom Boot	Syscom boot version
Panel CPU	Panel CPU firmware version
Touch Cnt	Touch Controller (ATMEL) version
eMMC	eMMC firmware version
Disc Unit Name	Disc mecha module name
Disc Version	Disc mecha module firmware version
Disc Region	DVD Region code

Item	Description
Disc CPRM	Disc mecha CPRM check sum
Disc Serial No	Disc mecha module serial no
Macrovision (Sys)	Macrovision version (System)
Macrovision (Disc)	Macrovision version (Disc mecha module)
HDCP Key	HDCP Key Condition (-/OK)
Navigation S/W	Navigation software Version (Displayed in DNX model only)
Base Map	Navigation database version (Base Map) (Displayed in DNX model only)
Detail Map	Navigation database version (Detail Map) (Displayed in DNX model only)
Junction View	Navigation database version (Junction View) (Displayed in DNX model only)
Dealer POI	Dealer POI Version (OEM model only) (Displayed in DNX model only)
DATA1	Opening data version
DATA2	Wallpaper version
DATA3	Steering Remote threshold version

After the version check is executed, items that are inconsistent with the version check are displayed in red.

Key	Operation
Export	Output version information as text file to USB memory.
Back	Return to the Production menu.

4.6.2 Sensor Status

Mode for displaying the sensor information entered into the system.

Sensor Status				Back
Gyro	:32812(0)	Gsens	: -	
Import Calib. Data		Elevation	: -	
Speed Pulse	: 0 ()	Dpp,Ofs,Sns	: 0, 2, 0	
G-Ofs	: 0, 0, 0	Dpp Learn Cnt	: 0	
Sns Learn Cnt	: 0	Ofs Learn Cnt	: 487	
Reverse Signal	: Off	Parking Brake	: On	
Illumination	: Off	Light Sensor	: 0	

Sensor Status				Back
GPS				
Satellites	: 0/0	Lat.	: S0-0-22	
HDOP	: 99	Lon.	: W0-0-16	
PDOP	: 99	Alt.	: 0	
-	: -	-	: -	
-	: -	-	: -	
-	: -	-	: -	

Item	Description
Gyro	Display the absolute value of gyro and offset value.

Item	Description
Gsens	Display the absolute value (x y z) of the acceleration sensor.
Import Calib. Data	Transition to the Gyro adjustment data import screen. *Group B is not displayed.
Elevation	Display the tilt angle.
Speed Pulse	Display vehicle speed pulse value.
DppOfsSns	Display learning level value. (Dpp Ofs Sns)
G-Ofs	Display the learning center value (x y z) of the acceleration sensor.
Dpp Learn Cnt	Display the number of learning of Dpp. (moving distance per vehicle speed)
Sns Learn Cnt	Display the number of learning of SNS (gyrosensitivity)
Ofs Learn Cnt	Display the number of learning of Ofs (gyro offset)
Reverse Signal	Display the state of the reverse input (On / Off)
Parking Brake	Display the state of the parking input (On / Off)
Illumination	Displays the illumination input status (On / Off)
Light Sensor	"0" fixed
Satellites	Display the GPS reception status. (number of tracking satellites) / (number of reception satellites)
Lat.	Display the GPS positioning latitude.
HDOP	Display the HDOP (horizontal accuracy reduction rate) of GPS.
Lon.	Display the GPS positioning longitude.
PDOP	Display the PDOP (position accuracy reduction rate) of GPS.
Alt.	Display the GPS positioning altitude (m).
-	Display the up to 32 information of acquired satellites.

Key	Operation
Back	Return to the Production menu.

4.6.2.1 How to import Gyro Adjustment Data

When DIGITAL IO PWB or GYRO PWB was replaced, it is necessary to import Gyro Adjustment Data following the instructions below.

- (1) Download the Adjustment data.
 - a) Open "Car MM Gyro Adjustment Data" on top CUSiS page.
 - b) Enter the PWB number, serial number and sub PWB number and then press the search button.
 - c) Click the Download button to download the Gyro Adjustment Data File.
- (2) Storage file
 - a) Unzip the zipped file downloaded and Rename the folder to "SENSOR_CALB".
File Name: "CALBDATA_****.bin"
 - b) Move the unzipped folder (including a file) to USB memory.

* Error message will be displayed when Several files are in SENSOR_CALB folder.

- (3) Connect the USB and get into Service menu.
- (4) Select "Sensor Status" in Service menu.

- (5) Select "Import Calib.Data" in Sensor Status and execute the transfer process.

Sensor Status

Gyro :32812(0) Gsens : -
 Import Calib. Data Elevation : -
 Speed Pulse : 0 () Dpp,Ofs,Sns : 0, 2, 0
 G-Ofs : 0, 0, 0 Dpp Learn Cnt : 0
 Sns Learn Cnt : 0 Ofs Learn Cnt : 487
 Reserse Signal : Off Parking Brake : On
 Illumination : Off Light Sensor : 0

Import Calib. Data

Processing...

Result OK

Import Calib. Data

OK

Result NG

Import Calib. Data

NG E01

NG is displayed with Error number and check the cause in errors

Error number	Error Details
E01	No target folder in USB mass storage
E02	No target file in USB mass storage
E03	There are multiple target files in the USB mass storage
E21	Read NG of target file
E22	Checksum NG of target file
E31	Writing NG of the target file contents mismatch of write data and data in USB mass storage.
E99	Errors that can not be classified as above

Key	Operation
Back	Return to the Sensor Status screen.

4.6.3 Error Log

Mode for displaying the Error Log of the system.

This displays the error number when an error occurred in the system and number of times the error occurred.

Click up and down to scroll the screen.

- (1) Select the "Error Log" on the Service Menu

Error Log

SoC Error

Total n : 0 E (n) : 00-00
 E (n-1) : 00-00
 E (n-2) : 00-00

Syscom Error

Total n : 0 E (n) : 00-00
 E (n-1) : 00-00

Error Log

Total n : 0 E (n) : 00-00
 E (n-1) : 00-00
 E (n-2) : 00-00

Connection Error

DC(Connect) :1 DC(Con Leak) :0
 DC(Shortcut) :1 DC(Other) :0

Item	Description
<SoC Error>	
Total	Total number of occurrences of SoC error
E(n)	SoC error: latest error code
E(n-1)	SoC error: one previous error code from latest
E(n-2)	SoC error: two previous error code from latest
<Syscom Error>	
Total	Total number of occurrences of Syscom Error
E(n)	Syscom Error: latest error code
E(n-1)	Syscom Error: one previous error code from latest
E(n-2)	Syscom Error: two previous error code from latest
<Connection Error>	
DC(Connect)	Total occurrence count of DC offset error (Abnormal connection)
DC(Con Leak)	Total occurrence count of DC offset error (Capacitor leak)
DC(Shortcut)	Total occurrence count of DC offset error (Battery short/GND short)
DC(Other)	Total occurrence count of DC offset error (Other)

Key	Operation
Back	Exit the Error Log screen

4.6.3.1 Soc error code

Display format: XX-EE

XX: Error module		XX: Error type	
01	SoC	00	Normal Operation
02	Panel CPU	01	Exception occurred
03	Syscom	02	Fatal error
04	SoC OS	03	App Hang Up
05	Navi App	04	Startup failure
06	Ent App	05	CPU hang
07	GUI	06	Insufficient memory
08	MSC App		
09	Sysman		
0A	Driver		
0B	Framework		

4.6.3.2 Syscom error code

MOD code	ERR code	Error Details
0x00	0x00	RAM error
	0x01	ROM abnormality
	0x02	Normal stack shortage
	0x03	Insufficient interrupt stack
	0x04	Heap shortage
	0x05	Message queue shortage
	0x06	Punctual processing overload
	0x07	Watchdog time-out
0x01	0x00	Safety timer time out (Main internal module is not ready for ACC OFF)
	0x01	Safety timer time out (Power-off request not received from Navi CPU)
0x02	0x00	NSIO transmission impossible
	0x01	NSIO unreceivable
	0x02	NSIO transmission queue shortage
	0x03	EBOOT startup failure
	0x04	Navi CPU startup failed
	0x07	Other error
0x03	0x00	Battery voltage
	0x01	Power amp temperature
	0x02	gyro
	0x03	G sensor
	0x04	Main microcomputer temperature
	0x05	Tilt voltage

MOD code	ERR code	Error Details
0x04	0x00	Buffer shortage
	0x01	Audio Tuner IC communication error
	0x02	FM Vics Tuner IC communication error
	0x03	1Seg IC communication error
	0x04	EEPROM communication error
	0x05	EVOL (TDA7715LV) communication (I2C communication) error
	0x06	Beep LSI (ML22Q394) communication (I2C communication) error
	0x07	Display I2C communication (ML86250) (I2C _ VIDEO) error
0x05	0x00	Mechanical malfunction
	0x01	Tilt voltage value
	0x07	Other error
0x06	0x00	DAB microcomputer communication error
	0x01	Cayman microcomputer communication error
	0x07	Other error
0x07	0x00	SIO communication disabled
	0x07	Other error
0x08	0x00	UART communication error
	0x01	SBAS can not be set
	0x07	Other error
0x09	0x00	SIO communication error
	0x07	Other error
0x0A	0x00	UART communication error
	0x07	Other error
0x0B	0x00	DARC decoder IC SIO Unable to transmit
	0x01	DARC decoder IC SIO unreceivable
	0x02	S meter error
	0x07	Other error
0x0C	0x00	UART communication error
	0x07	Other error
0x0D	0x00	UART communication error
	0x01	No camera response T.O.
	0x02	Incomplete reception T.O.
	0x03	Settings saved for more than 11 times
	0x07	Other error
0x0E		* Definition only
0x0F	0x00	UART communication error
	0x07	Other error
0x10	0x00	PANEL (LAPIS) (I2C communication) error
	0x01	I2C communication error of FERRSCALE
	0x02	ASEL I2C communication error
0x11	0x00	LINE_MUTE voltage abnormality
	0x07	Other errors (Around view camera voltage abnormality etc.)

MOD code	ERR code	Error Details
0x12	0x00	DSP communication error
	0x07	Other error
0x13	0x00	ADS communication error
0x1D	0x00	Initialization table transmission failure
	0x01	Tuning table transmission failed
	0x07	Other error
0x1E	0x00	UART communication error
	0x01	ModuleCfgInd reception error
	0x07	Other error
0x1F	0x00	Area data size mismatch
	0x01	Parity / Sam mismatch
	0x07	Other error

Upper 5 bits: MOD cord, lower 4 bits: Error code

(ex) Watchdog timeout

bit	7	6	5	4	3	2	1	0		
value	0	0	0	0	1	1	0	1	ERR0	Error code
									ERR1	
									ERR2	
									MOD0	MOD code
									MOD1	
									MOD2	
									MOD3	
									MOD4	

4.6.4 Software update

4.6.4.1 System Update

4.6.4.1.1 All update

This is the standard update method. This method is also used to update the firmware for user.

- (1) Save the update package in the root directory of the USB memory.

*The update package has a predetermined folder structure.

Example: \S_V1_0_0089_0100
+ BOOT_V1_1_0019_0100
+-- (update file)
+-- (update file)
:
+ MAIN_V1_0_0019_0100
:
+ SOC_V1_1_0039_0100
:

- (2) Connect the USB memory while the power of the unit is turned on.
- (3) A confirmation dialog box for the update appears, press "Yes" to start updating.

Caution:

- Do not turn off the power during update.
- Do not save other update packages and files for individual updates in the USB memory.
- This method cannot be downgraded.

4.6.4.1.2 Individually update

Each CPU firmware can be updated individually from the Service Menu.

This method can be downgraded as it does not perform version check.

- (1) Save the updata in the root directory of the USB memory.

Example:

\SOC_V0_1_1029_1100
+-- Soc_V0_1_1029_1100.nfu
+-- SumList.ini

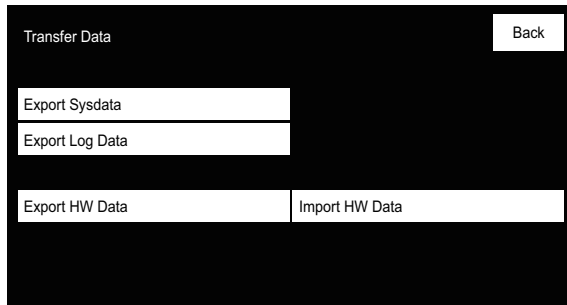
- (2) Connect the USB memory while the power of the unit is turned on.
- (3) Displays the service menu, and select the "Update".
- (4) Select the CPU to be updated.
- (5) A confirmation dialog box for the update appears, press "Yes" to start updating.

Caution:

Do not turn off the power during update.

4.6.5 Transfer Data

Mode for transferring the data in the system.



Key	Operation
Export Sysdata	Transfer the sysdata in the system to USB memory
Export Log Data	Transfer the log data in the system to USB memory
Export HW Data	Transfer the HW Config data in the system to USB memory
Import HW Data	Transfer the HW Config data in the USB memory to system
Back	Exit the Transfer Data screen

4.6.5.1 Transfer of Sysdata

Connect the USB memory in advance.

- (1) Select the "Transfer Data" in the service menu, and Displays the Transfer Data screen.
- (2) Select the "Export Sysdata".
- (3) A confirmation dialog box appears, select "Yes" to transfer the data.

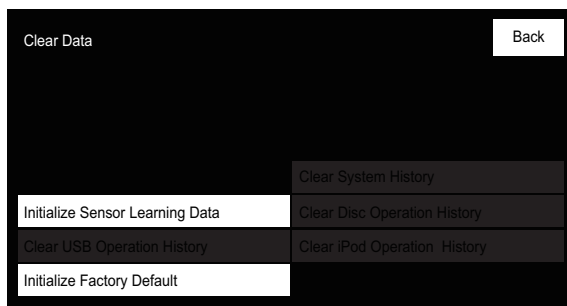
4.6.5.2 Transfer of Log Data

Connect the USB memory in advance.

- (1) Select the "Transfer Data" in the service menu, and Displays the Transfer Data screen.
- (2) Select the "Export Log Data".
- (3) A confirmation dialog box appears, select "Yes" to transfer the data.

4.6.6 Clear Data

Mode for clearing the data in the system.



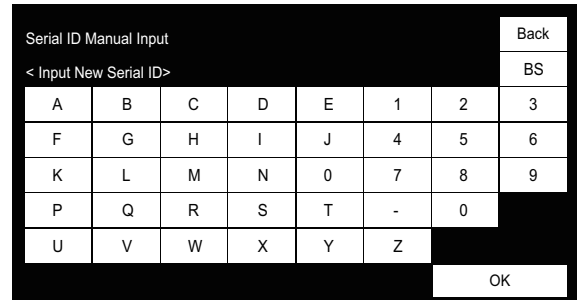
Key	Operation
Initialize Sensor Learning Data	Initialize learning data of sensor (gyro sensitivity, moving distance per pulse speed of vehicle, gyro offset)
Initialize Factory Default	For initializing to factory default. *Clear the Personal identification number/DC offset number of errors together
Back	Exit the For clearing to Data screen

4.6.6.1 Execution method

- (1) Select the "Clear Data" in the service menu, and displaying the Clear Data screen.
- (2) Select the "Initialize Factory Default".
- (3) A confirmation dialog box appears, select "Yes" to clear.

4.6.7 Serial ID Input

Mode for entering the serial number recorded in the system manually.



Key	Operation
A to Z, -, 0 to 9	Input the characters
BS	Deletes one character from the characters entered. * Press and hold to delete all characters.
OK	Confirms/saves the characters entered.
Back	Exit the Serial ID Manual Input screen

4.6.7.1 Changing method

- (1) Select the "Serial ID Manual Input" from the service menu, and displaying the Serial ID Manual Input screen.
- (2) Use the A to Z, 0 to 9 keys to enter the serial number (8 digits) to be saved.
- (3) The OK button is enabled once 8 digits have been entered; press the OK button to save the serial number.

*When press the OK button with serial number not yet entered, the recorded serial number is deleted.

4.6.8 Y No Input (DNR model only)

Mode for entering the model number (Y No) recorded in the system manually.

Y No Manual Input							Back
< Input New Y No>							BS
A	B	C	D	E	1	2	3
F	G	H	I	J	4	5	6
K	L	M	N	0	7	8	9
P	Q	R	S	T	-	0	
U	V	W	X	Y	Z		
							OK

Key	Operation
A to Z, -, 0 to 9	Input the characters
BS	Deletes one character from the characters entered. * Press and hold to delete all characters.
OK	Confirms/saves the characters entered.
Back	Exit the Y No Manual Input screen

4.6.8.1 Changing method

- (1) Select the "Serial ID Manual Input" from the service menu, and displaying the Serial ID Manual Input screen.
- (2) Use the A to Z, 0 to 9 keys to enter the model number (Y No. 9 digit) to be saved.
- (3) The "OK" button is enabled once 9 digits have been entered; press the "OK" button to save the model number.

4.6.9 LCD Adjust

Mode for adjusting the image quality in the production line.

4.6.9.1 Flicker adjustment

- (1) When "LCD Adjust" is selected in the Production Menu, transition to the flicker adjustment screen.
- (2) Select the key from "Vcom 1" to "Vcom 8" (8 options) to seek out the Flicker amplitude level is minimum condition. (Select best screen condition)

LCD VCOM Adjust								Back
Vcom 1	Vcom 2	Vcom 3	Vcom 4	Vcom 5	Vcom 6	Vcom 7	Vcom 8	
								OK

- (3) Press "OK" to memory this condition and then move to "LCD White Balance Adjust" mode, automatically.

Key	Operation
Vcom1	Set Vcom1
Vcom2	Set Vcom2
Vcom3	Set Vcom3
Vcom4	Set Vcom4
Vcom5	Set Vcom5
Vcom6	Set Vcom6
Vcom7	Set Vcom7
Vcom8	Set Vcom8

Key	Operation
Up	Switch Vcom1 → Vcom2 → ---
Down	Switch Vcom8 → Vcom7 → ---
OK	Store
Back	Return to to Production Menu

4.6.9.2 White balance adjustment

- (1) After finished Flicker adjustment, automatically change to this mode.
- (2) Change the adjustment values so that the screen is most white.
- (3) Press "OK" to memory the adjusted value.

LCD White Balance Adjust				Back
B	-1	XXX	+1	
	-10		+10	
R	-1	XXX	+1	
	-10		+10	
G	-1	XXX	+1	
	-10		+10	
				OK

After saving the data by pressing the OK button, transition to the Production Menu screen.

Key	Operation
Key	Operation
B +1	Offset Gain B Gain Up
B -1	Offset Gain B Gain Down
B +10	Offset Gain B Gain Up +10
B -10	Offset Gain B Gain Down -10
R +1	Offset Gain R Gain Up
R -1	Offset Gain R Gain Down
R +10	Offset Gain R Gain Up +10
R -10	Offset Gain R Gain Down -10
G +1	Offset Gain G Gain Up
G -1	Offset Gain G Gain Down
G +10	Offset Gain G Gain Up +10
G -10	Offset Gain G Gain Down -10
OK	Store
Back	Return to to Production Menu

The font color of the "LCD Adjust" button on the first page of the Production Menu changes depending on the result.

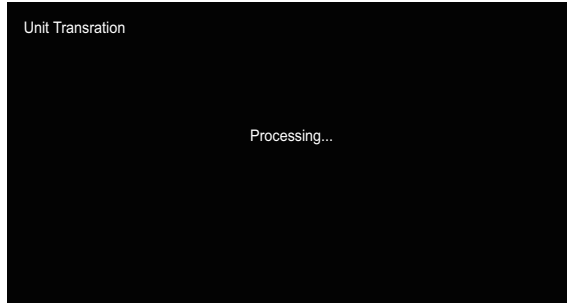
- Not performed: Black
- OK: Blue
- NG: Red

4.6.10 UnitID Translation (DNR model only)

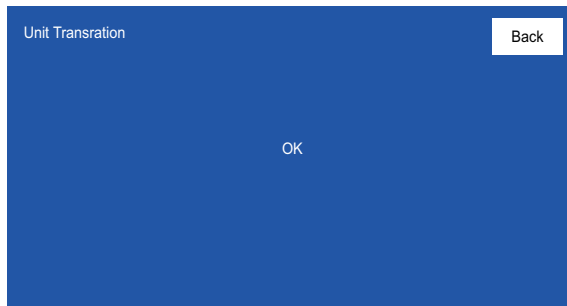
Mode for writing the unit ID/unlock data.

- (1) Download the Unit ID data (zip file) from the CUSiS Garmin Unit ID Download site.
- (2) Unzip the file and copy each folder to the USB root directory.
- (3) Connect the USB memory.
- (4) Select "Unit ID Translation" from the service menu to process the transfer.

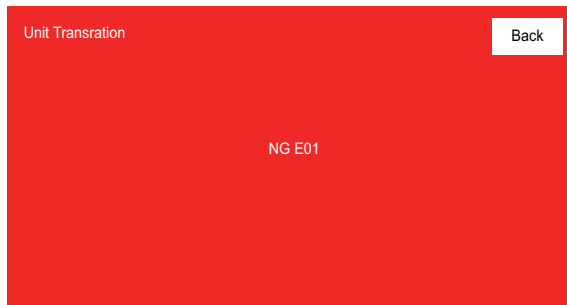
Checking...



Result OK



Result NG



The error description corresponding to the error number is as follows.

Error number	Description
E01	There is no file within USB mass storage
E02	There is no file within USB mass storage (gmap-suppl.gma)
E03	There is no file within USB mass storage (gmap-suppl.unl)
E04	There is no file within USB mass storage (gtm_nvml.dat)
E05	There is no file within USB mass storage (CSV)
E11	Serial number read NG from the system
E12	Y number read NG from the system
E21	Can not search the supported Unit ID (not found)
E22	CSV file parsing error
E31	UnitID writing NG to the system

Error number	Description
E32	Copy NG to gmapsuppl.gma/temp/Garmin
E33	Copy NG to gmapsuppl.gma/Syslog/Garmin
E34	Copy NG to gmapsuppl.unl/temp/Garmin
E35	Copy NG to gmapsuppl.unl/Syslog/Garmin
E36	Copy NG to gtm_nvml.dat/temp/Garmin
E37	Copy NG to gtm_nvml.dat/Syslog/Garmin
E99	Errors that can not be classified in the above

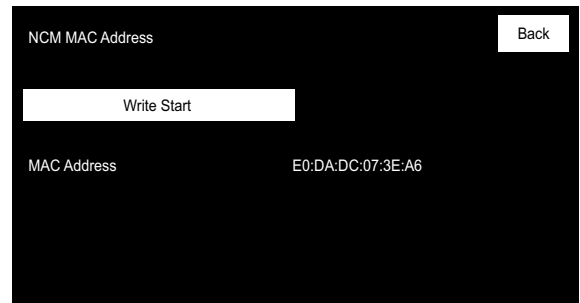
4.6.11 NCM MAC Address

Mode for writing of Apple CarPlay USB NCM MAC address and view current MAC address.

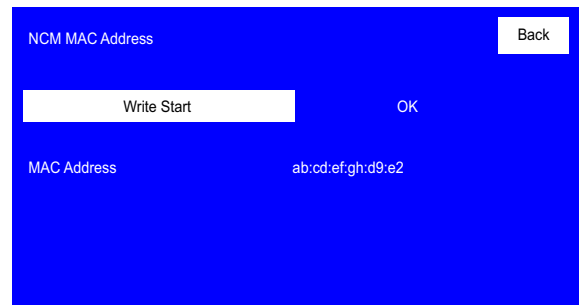
When DIGITAL IO PWB was replaced, it is necessary to writing MAC address following the instructions below.

- (1) Download the MAC address data.
 - a) Open "NCM MAC Address Data" on top CUSiS page.
 - b) Enter the Y number and serial number.
 - c) Click the Download button to download the MAC address data File.
- (2) Storage file
 - a) Unzip the zipped file downloaded.
 - b) Move the unzipped folder (including a file) to USB memory.

*Error message will be displayed when Several files are in folder.
- (3) Connect the USB memory containing the MAC address data.
- (4) Select "Write Start" from the service menu to process the transfer.



Result OK



Result NG



Item	Description
MAC Address	Display the current MAC address.

Key	Operation
Write Start	Start MAC address write processing.
Back	Return to the Service menu.

4.6.12 Initialize Factory Default (for Service Menu)

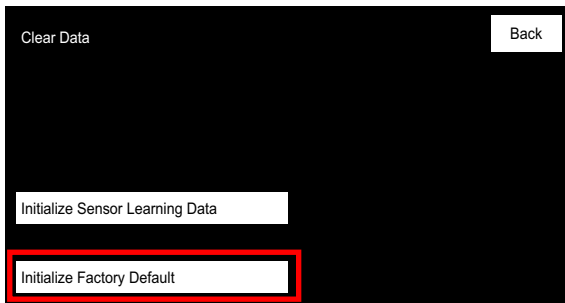
Mode for initializing to factory default.

Initialize all of the backup memory of the system.

The flag for changing the font color of the button according to the OK/NG status in the Production Menu is also cleared.

The following data are not initialized.

- LCD adjustment value
 - Panel mecha adjustment value
 - Key illumi adjustment value
 - Gesture sensor adjustment value
 - Model number / serial ID
 - Region code
 - Tuner Area Setting
 - Opening customize data
 - Wallpaper customize data
 - Initial value customize data
 - Learning remote control customize data
- (1) Select the "Clear Data" in the Service Menu.
 - (2) Select the "Initialize Factory Default" in the Clear Data screen.



- (3) When the "Yes" in the confirmation dialog selected, Perform the factory initialize.
- (4) After the initialization is complete, press OK to restart the unit.

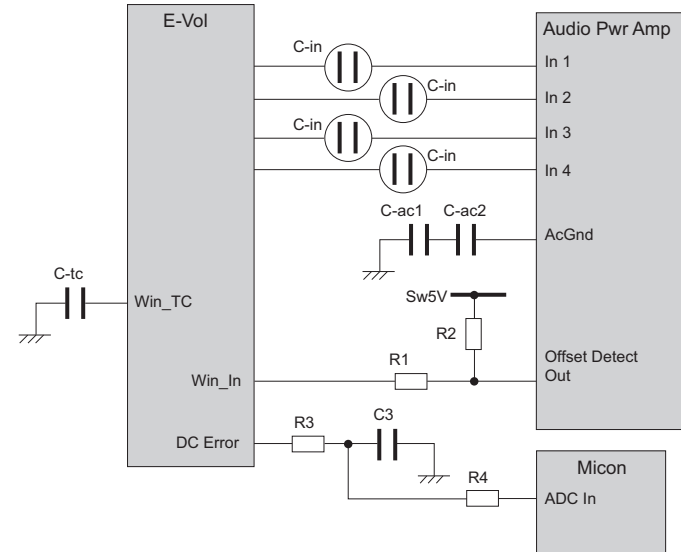
4.7 DC Offset error description

4.7.1 DC Offset detection circuit design

- Purpose:
To prevent breakdown, when occur DC offset between speaker output "+" and "-".
- Target:
Detect DC offset, then stop the Power Amp operation and shift to specified condition.

4.7.2 Possible causes of DC offset at speaker output lines

- (1) Mis-connection for Speaker output for example touch to car body or battery line.
- (2) Current leak of coupling capacitor for Power IC input.
- (3) Current leak of AC-GND capacitor for Power IC AC-GND.
- (4) Capacitor shorted of above parts due to foreign object.

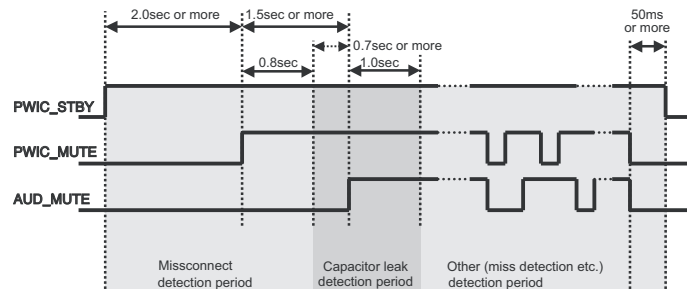


4.7.3 Type of checking

4.7.3.1 To detect DC Offset Error

- Mis-connection
 - Short any one speaker out line to GND or Vcc
- Capacitor leak
 - Parallel 330kΩ to either any one of coupling cap or AC-GND capacitor (to simulate current leakage of capacitor)
 - Shorted either any one of coupling capacitor or AC-GND capacitor.

4.7.4 Detection Timing chart



4.7.5 Manipulate after detect DC Offset

- If detected error 10 consecutive times, and 10th error occurred in "Mis-connect detection period", judge as "Mis-connect".
- If detected error 10 consecutive times, and 10th error occurred in "Capacitor leak detection period", judge as "Capacitor leak".
- If detected error 10 consecutive times, and 10th error occurred in "Other detection period" and detected another 10 errors consecutively, then judge as "Other".
- If judge as "Mis-connect".
 - turn off speaker output.
 - display "MISWIRING", check wiring connection then reset.
 - key access disable except button of Eject, Reset and service mode
 - record error in EEPROM "DC1 ERR"
 - the product is able to be recovered by Reset button.
- If judge as "Capacitor leak".
 - turn off speaker output.
 - display "WIRING", check wiring connection then reset.
 - key access disable except button of Eject, Reset and service mode
 - record error in EEPROM "DC2 #" (# means counter number)
 - the product can be recovered by pressing the Reset button before the capacitor leak error counter reach "DC2 4".
After that, only clear the counter back to "0" can recover the product.
- If judge as "Other" (manipulation same as mis-connect)

4.7.6 How to cancel DC offset error

Check to be sure the terminals of the speaker leads are covered with insulating tape properly, then reset the product.

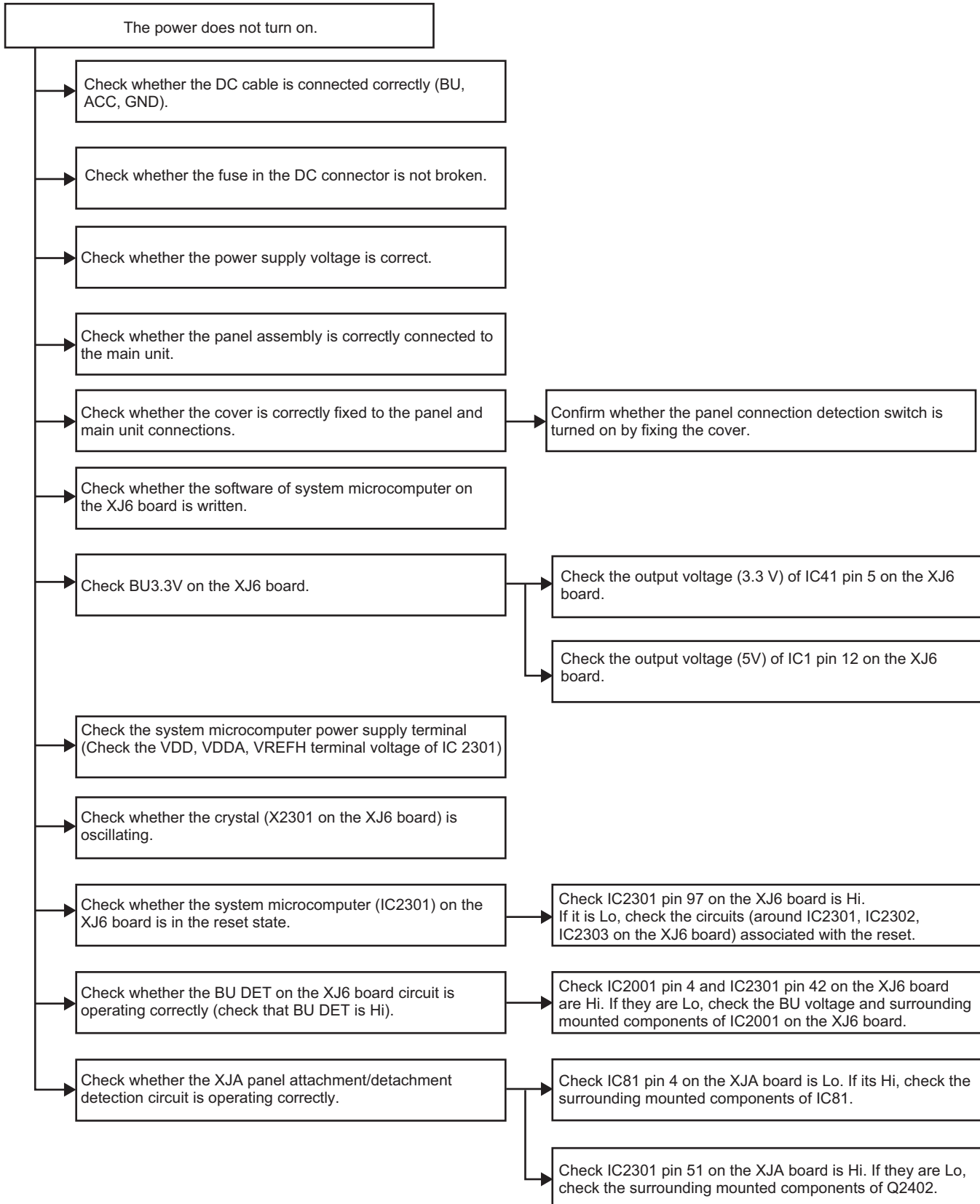
If an error message does not disappear even after a reset, there is a need for internal repair.

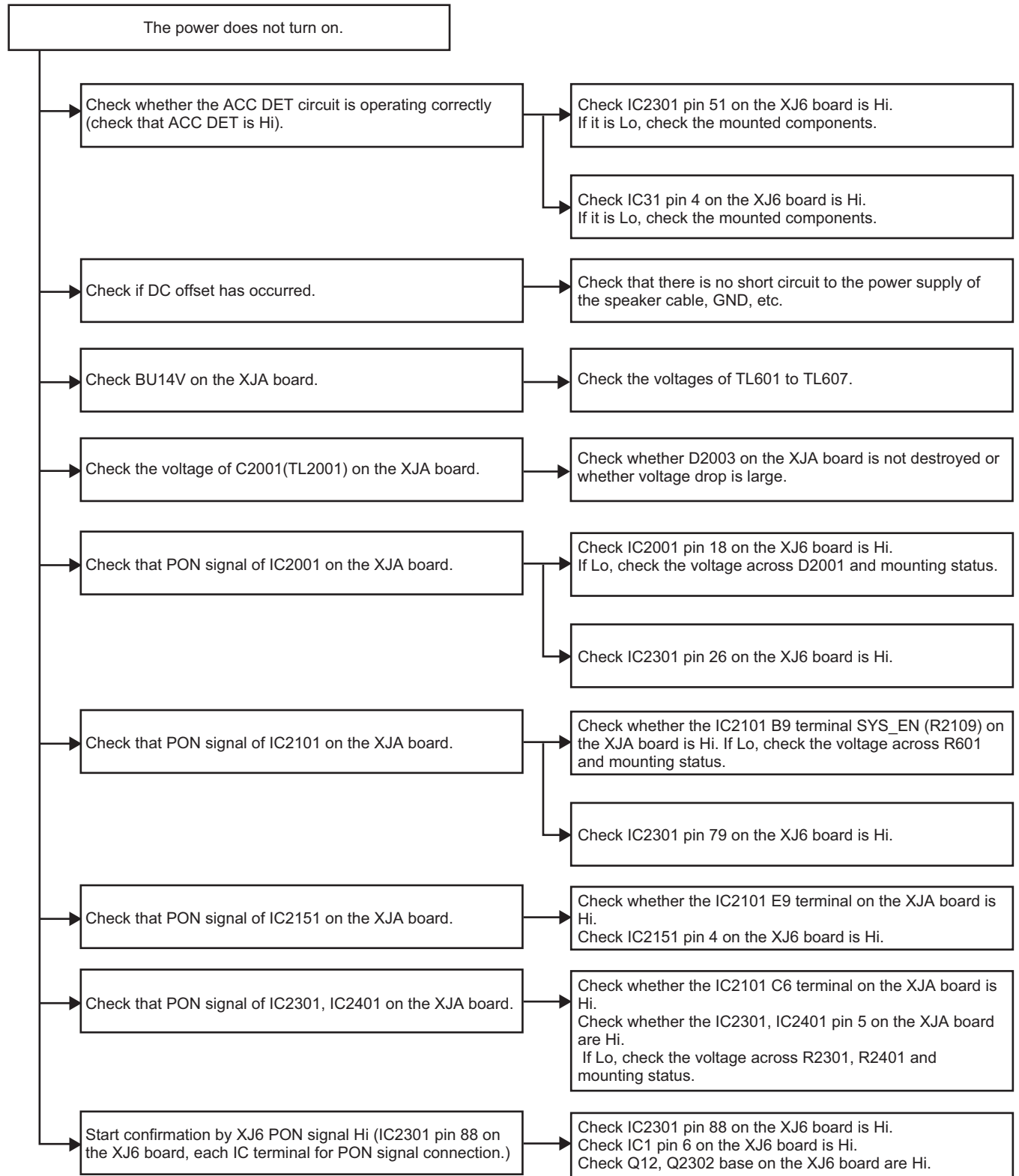
4.7.7 How to clear DC offset error recorded in EEPROM

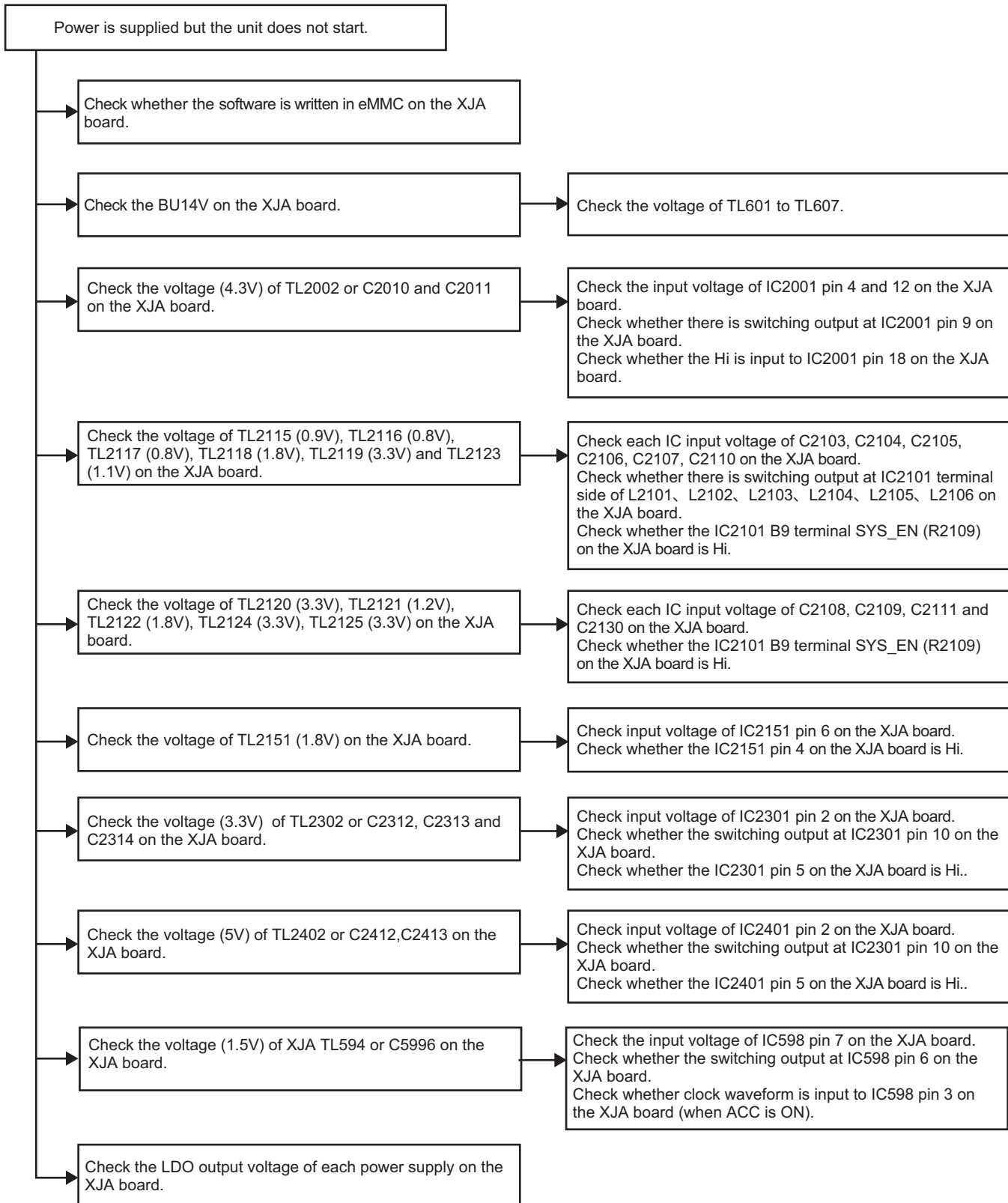
Refer to "4.6.6 Clear Data".

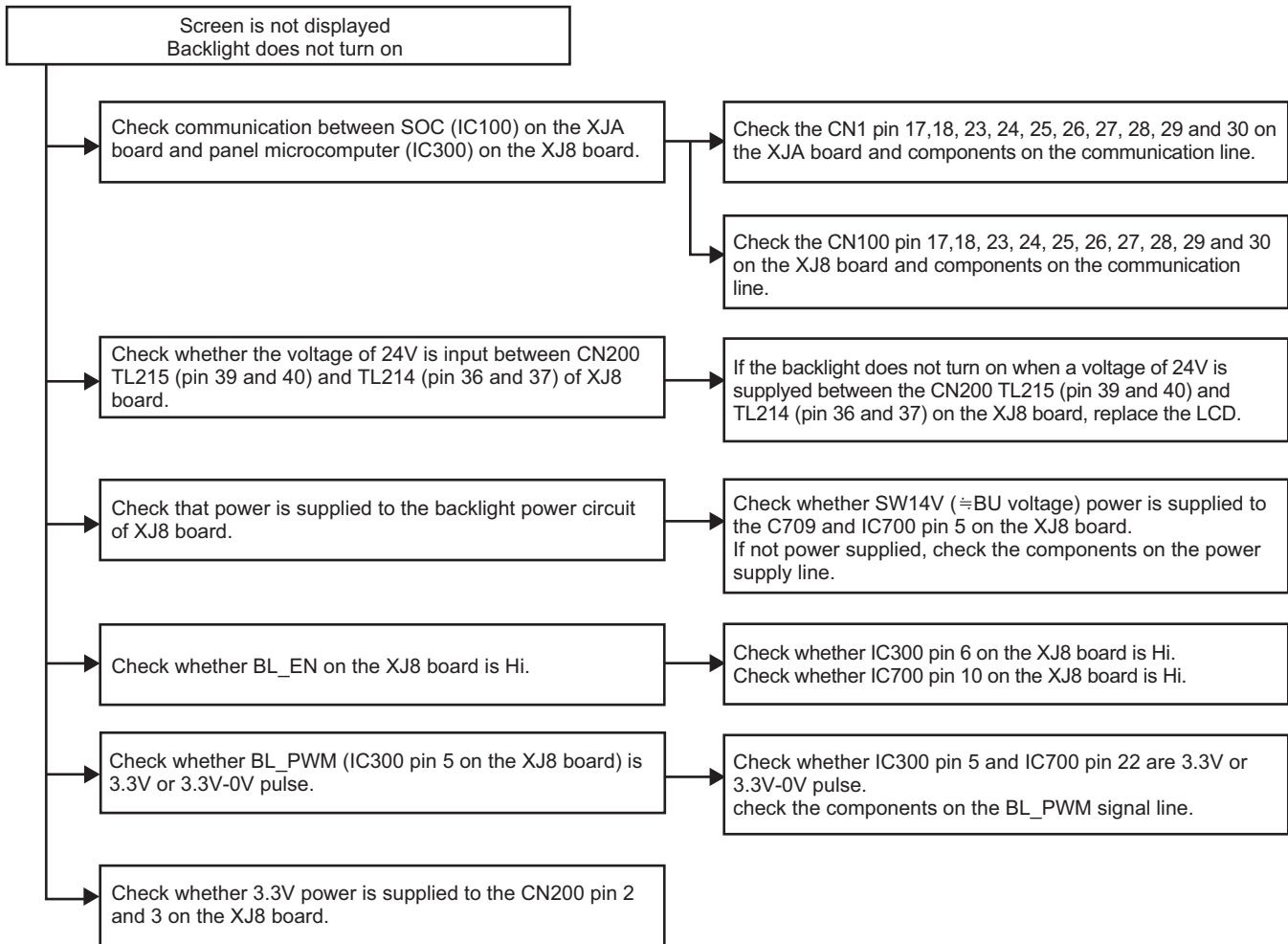
SECTION 5 TROUBLESHOOTING

5.1 Analysis Procedure When the Power Does Not Turn On











KENWOOD

JVCKENWOOD Corporation

Automotive 2967-3, Ishikawa-machi, Hachioji-shi, Tokyo, 192-8525, Japan

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